

Finite-Sample Welch–Satterthwaite Inference for Comparing Mutual Information Between Independent Categorical Populations

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Declaration

I certify that this thesis is my own work and that all sources used in its preparation have been acknowledged. Material derived from collaborative work, software, and prior research is identified where it appears. This thesis has not been submitted for another degree except where permitted by the relevant University regulations.

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Abstract

Mutual information measures statistical dependence without restricting that dependence to a linear or monotone form. Standard significance procedures for categorical mutual information primarily test whether two variables are independent within one population. This thesis studies a different problem: whether two independent populations have equal mutual information, $H_0 : I(P) = I(Q)$, while allowing their complete joint distributions to differ.

The usual analytic approach forms a bias-corrected difference of plug-in mutual information estimates, divides it by an influence-function standard error, and compares the resulting statistic with a standard normal distribution. This normal Wald reference does not represent finite-sample uncertainty in the estimated variance. The thesis develops an expanded Welch–Satterthwaite correction that retains the same effect estimate and standard error but derives mutual-information-specific component degrees of freedom from the sampling variability of the complete variance functional. The resulting procedure is deterministic and requires $O(rc)$ operations for an $r \times c$ table.

The method was evaluated using 60 fixed pairs of positive-support categorical populations with exactly equal mutual information. The experiment covered six table shapes, five sampling regimes, sample sizes from 50 to 1,000, and 10,000 independent table-pair replicates per population pair. At nominal $\alpha = 0.05$, mean absolute calibration error across all scenarios was 0.01342 for normal Wald, 0.01194 for simple Welch–Satterthwaite, and 0.00825 for expanded Welch–Satterthwaite. The expanded method left the well-sampled control essentially unchanged and reduced error in the skewed, rare-cell, and widespread-sparsity regimes, although it did not restore exact calibration in the most difficult cases. Its average power loss relative to normal Wald was approximately one percentage point, and median runtime remained below 0.15 milliseconds for the tested tables.

These results support expanded Welch–Satterthwaite inference as a low-cost finite-sample correction for comparing positive mutual information values. The method is not a universal sparse-table solution and does not apply at the nonregular

independence boundary, where the first-order mutual-information variance vanishes.

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Chapter 1

Introduction

1.1 Comparing dependence across populations

Many empirical studies ask whether the relationship between two variables changes across populations. A treatment may alter the association between a biomarker and an outcome; environmental conditions may change the dependence between two species; or two periods may exhibit different coupling between components of a complex system. In each case, the quantity of interest is the strength of association, not only the marginal behaviour of either variable.

Mutual information (MI) provides a general measure of statistical dependence ([Shannon, 1948](#)). For categorical variables, it compares the observed joint distribution with the joint distribution that would arise under independence. Unlike correlation, MI does not restrict dependence to a linear or monotone form. This generality makes MI useful in complex-systems research and in applications where the structure of an association is not known in advance ([Brillinger, 2004](#); [Lizier, 2014](#)).

The most familiar significance test for categorical MI asks whether two variables are independent within one population. The corresponding null is $I(X; Y) = 0$, and the likelihood-ratio statistic has an asymptotic chi-squared distribution under standard fixed-alphabet conditions. A permutation test can address the same question by shuffling one variable and thereby destroying its association with the other.

Comparing dependence across two populations is a different inferential problem. Let P and Q denote the joint distributions of the same aligned categorical variables in two independent populations. This thesis studies

$$H_0 : I(P) = I(Q) \quad \text{against} \quad H_1 : I(P) \neq I(Q). \quad (1.1)$$

The complete distributions may differ under this null. Their margins, cell probabilities, and patterns of association need not be equal. Equation (1.1) therefore defines a weak null about one scalar functional of each distribution.

This distinction determines the appropriate analysis. Applying a separate chi-squared independence test to each table does not test whether the two MI values are equal. Shuffling one variable within either population also targets independence rather than Equation (1.1). The required method must estimate the signed difference $I(P) - I(Q)$ and quantify its sampling uncertainty.

1.2 The finite-sample problem

For an $r \times c$ contingency table, the plug-in MI estimator replaces each population probability by its observed proportion. The estimator is simple and consistent, but it is nonlinear and has positive finite-sample bias (Miller, 1955; Moddemeijer, 1989; Paninski, 2003). The leading bias depends on the alphabet dimensions and sample size. Consequently, two groups with equal population MI can have differently biased estimates when their sample sizes differ.

Away from independence, MI is a smooth functional of the cell probabilities. A first-order Taylor expansion expresses the estimation error as an average of centred pointwise-MI contributions. This representation gives an analytic standard error and leads to a normal Wald test for Equation (1.1). The calculation is deterministic and requires one pass over each table.

The normal Wald test treats the estimated standard error as sufficiently stable. That approximation becomes less reliable when the sample is small, the margins are skewed, or many cells have low expected counts. In these settings, repeated samples can produce substantially different estimates of the MI variance. Ignoring that second layer of uncertainty can make the normal-reference test reject too often.

Welch's unequal-variance test addresses an analogous issue when comparing two means. It forms a Student statistic and uses effective degrees of freedom that depend on the two estimated variance contributions (Satterthwaite, 1946; Welch, 1947). Hutcheson transported this architecture to the comparison of Shannon diversity values (Hutcheson, 1970). A direct transport to MI, however, does not fully describe the uncertainty of the MI variance estimate. Changing one observation alters the joint probabilities, the margins, every affected pointwise-MI value, and the estimated MI variance.

1.3 Aim and research questions

This thesis develops and evaluates an MI-specific Welch–Satterthwaite reference for the weak null in Equation (1.1). The proposed test keeps the bias-corrected MI difference and first-order standard error of the normal Wald test. It changes only the reference distribution used to calibrate the standardized statistic.

Figure 1.1 gives the high-level calculation. The two tables are processed separately until their bias-corrected MI estimates and variance contributions are combined. The proposed contribution lies in the final Welch-type calibration: its degrees of freedom are derived from the sampling stability of the complete MI variance estimator.

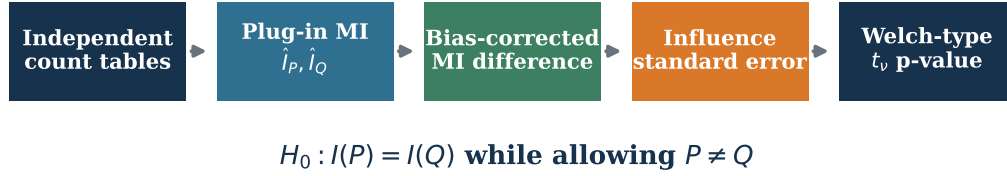


Figure 1.1: High-level analytic pipeline for comparing MI between two independent count tables. Expanded Welch uses the same effect estimate and standard error as the two baselines but supplies MI-specific component degrees of freedom for the final Student reference.

The study addresses three research questions.

RQ1: Method construction. How can the Welch–Satterthwaite principle be adapted to the nonlinear plug-in MI variance estimator?

RQ2: Statistical performance. Does the expanded method improve type-I error calibration and confidence interval coverage relative to normal Wald and simple Welch–Satterthwaite inference across well-sampled, skewed, sparse, and unequal-sample regimes?

RQ3: Practical trade-offs. What power, computational cost, validity-rate, and operating-range trade-offs accompany the correction?

1.4 Contributions

The thesis makes four contributions.

First, it derives component degrees of freedom for the estimated first-order MI variance. The derivation treats that variance as a complete functional of the joint distribution, including the dependence of pointwise MI on the row and column margins.

Second, it combines these component degrees of freedom into a deterministic expanded Welch–Satterthwaite test. The complete calculation requires $O(rc)$ operations for an $r \times c$ table and does not require resampling.

Third, it evaluates normal Wald, simple Welch–Satterthwaite, and expanded Welch–Satterthwaite inference on a common grid of exactly equal-MI population pairs. The grid separates well-sampled controls from skewed, rare-cell, ultra-sparse, and widespread-sparsity regimes.

Fourth, it characterises the method’s operating range. The results identify regimes in which the correction improves calibration, quantify its power and runtime costs, and show where first-order inference remains unreliable.

The contribution is deliberately bounded. Welch–Satterthwaite moment matching, influence-function theory, bias correction for plug-in information measures, and comparisons of MI-related indices all have established literatures. The new element is the MI-specific variance-influence degrees-of-freedom construction and its controlled finite-sample evaluation for the equal-MI weak null.

1.5 Scope

The analysis assumes two independent samples of aligned categorical pairs. The principal theory concerns fixed finite alphabets, positive population cell probabilities, and positive first-order MI variances. The target MI values in the primary validation are positive and bounded away from zero.

Testing independence is outside the method’s regular operating range. At $I(P) = 0$, the first-order MI influence function is zero and the usual normal expansion degenerates. The chi-squared independence limit then arises from a second-order expansion. Appendix D derives this boundary explicitly.

The thesis does not develop paired-sample inference, conditional MI, transfer entropy, or a universal sparse-table correction. These extensions require additional dependence structures or different asymptotic arguments.

1.6 Thesis structure

Chapter 2 introduces the information-theoretic and statistical foundations, reviews prior comparison methods, and identifies the specific methodological gap. Chapter 3 defines the two-sample problem and derives the shared bias-corrected statistic, normal Wald baseline, and simple Welch baseline. Chapter 4 derives the expanded MI-specific degrees of freedom.

Chapter 5 presents the population construction, simulation grid, evaluation metrics, and reproducibility controls. Chapter 6 reports calibration, coverage, power, validity, and runtime. Chapter 7 interprets the findings, defines the operating range, and discusses limitations. Chapter 8 answers the research questions and concludes the thesis.

Chapter 2

Background and Related Work

This chapter develops the concepts needed to compare MI between two categorical populations. It begins with entropy, pointwise MI, and population MI. It then reviews plug-in estimation, first-order inference, the Welch–Satterthwaite approximation, and prior comparisons of information quantities. The final section identifies the gap addressed by the expanded method.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is

$$h_X(x) = -\log p_X(x). \quad (2.1)$$

Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = -\sum_x p_X(x) \log p_X(x). \quad (2.2)$$

Entropy summarises the uncertainty of one random variable before its value is observed ([Shannon, 1948](#)).

For a pair (X, Y) , the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X | Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y .

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \dots, r\}$ and $Y \in \{1, \dots, c\}$. Write

$$p_{i+} = \sum_{j=1}^c p_{ij}, \quad p_{+j} = \sum_{i=1}^r p_{ij} \quad (2.3)$$

for the row and column margins. The pointwise mutual information associated with cell (i, j) is

$$\ell_P(i, j) = \log \left(\frac{p_{ij}}{p_{i+}p_{+j}} \right). \quad (2.4)$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts; a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values:

$$I(P) = \sum_{i=1}^r \sum_{j=1}^c p_{ij} \ell_P(i, j). \quad (2.5)$$

Equivalently,

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (2.6)$$

The quantity is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell with positive support.

Equation (2.5) is also the Kullback–Leibler divergence between the joint distribution and the product of its margins. MI therefore measures the overall departure from independence rather than one particular shape of association.

This scalar summary deliberately discards much of the information in the joint distribution. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. This distinction is central to the present test: the null fixes one scalar functional while leaving the remaining features of each population unrestricted.

The logarithm base determines the unit but not the inferential problem. If base two is used, every pointwise value and MI value is divided by $\log 2$. The standard error is divided by the same constant, so a studentized statistic is unchanged. This thesis uses natural logarithms and reports MI in nats.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij} . The empirical cell probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}}{n}. \quad (2.7)$$

Substitution into Equation (2.5) gives the plug-in estimator

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right). \quad (2.8)$$

The zero-cell convention follows from $\lim_{x \downarrow 0} x \log x = 0$. Empty cells contribute no direct term, but they still affect the empirical margins and the finite-sample behaviour of the estimator.

Under multinomial sampling, the entire table is random subject to its fixed total. Cell counts are dependent because increasing one count leaves fewer observations for the others. The plug-in estimator remains consistent for a fixed alphabet, but its finite-sample distribution depends on the complete probability table. Equal alphabet sizes and sample sizes do not imply equal sampling behaviour when one population has much rarer cells.

The plug-in estimator is upward biased. Finite sampling makes an empirical joint table appear more structured than its population distribution, which increases the estimated departure from independence. For a fixed positive $r \times c$ support, the leading bias is

$$\text{Bias}\{\hat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \quad (2.9)$$

This result follows from the corresponding leading entropy biases and is closely related to Miller's correction (Miller, 1955; Moddemeijer, 1989). More complete analyses show that sparse sampling and unknown support can produce substantial higher-order error beyond Equation (2.9) (Paninski, 2003).

The leading term can be seen from the entropy identity in Equation (2.6). For a discrete variable with m positive categories, the plug-in entropy has leading bias $-(m-1)/(2n)$. Applying this result to $H(X) + H(Y) - H(X, Y)$ gives

$$\text{Bias}\{\hat{I}(P)\} \approx -\frac{r-1}{2n} - \frac{c-1}{2n} + \frac{rc-1}{2n} \quad (2.10)$$

$$= \frac{(r-1)(c-1)}{2n}. \quad (2.11)$$

The correction is positive because estimating the larger joint distribution introduces more entropy bias than estimating the two margins.

Bias matters directly when comparing two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2} \left(\frac{1}{n_P} - \frac{1}{n_Q} \right). \quad (2.12)$$

Removing the leading term is therefore part of the effect estimate rather than an optional refinement of its reference distribution.

2.3 First-order inference for nonlinear functionals

MI is a nonlinear function of the cell probabilities. When all population cells are positive and the empirical table is close to its population value, a first-order Taylor expansion replaces this nonlinear function locally by a linear combination of the cell-probability errors. The resulting expansion is often expressed through an influence function ([Kandasamy et al., 2015](#)).

For MI, the observation-level influence function is centred pointwise MI:

$$\psi_P(i, j) = \ell_P(i, j) - I(P). \quad (2.13)$$

It has population mean zero. Its variance is

$$V(P) = \text{Var}_P\{\psi_P(X, Y)\} = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (2.14)$$

The plug-in MI error is, to first order, the sample mean of these influence values. Consequently,

$$\text{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n}. \quad (2.15)$$

This representation has two roles. First, it makes the central limit theorem available because the leading estimation error is an average of independent, mean-zero contributions. Second, it identifies the observation-level quantity whose variance determines the standard error. The raw categories themselves do not have a numerical variance relevant to MI; their effects on the MI functional are represented by centred pointwise MI.

The expansion is local. It describes sampling fluctuations near a fixed positive-support population and does not assert that the plug-in estimator is normally distributed for every finite table. Its accuracy depends on the sample size relative to the probability structure, not only on the total number of observations.

The central limit theorem then gives a normal approximation when $V(P) > 0$.

At exact independence, every pointwise-MI value is zero and $V(P) = 0$. The first-order term disappears, so a second-order expansion is required. This is why the chi-squared limit for testing independence and the normal limit for comparing positive MI values are not competing descriptions of the same regular case.

2.4 Wald and Welch–Satterthwaite inference

2.4.1 Normal Wald inference

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with its asymptotic normal reference. For two independent MI estimates, the first-order variance contributions add:

$$\text{Var}\{\hat{I}(P) - \hat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \quad (2.16)$$

Replacing the population quantities by estimates produces the normal Wald test developed formally in Chapter 3.

Wald inference is asymptotically valid under regularity conditions, but its finite-sample calibration depends on both the quality of the linear expansion and the stability of the estimated standard error. A normal reference does not separately represent the uncertainty of the two estimated variance contributions.

2.4.2 The Welch–Satterthwaite principle

Welch’s test compares two means when their population variances may differ (Welch, 1947). The denominator contains two independently estimated variance contributions. Satterthwaite’s approximation represents their sum by a scaled chi-squared variable with matching first two moments (Satterthwaite, 1946). The resulting effective degrees of freedom are

$$\nu = \frac{(\sum_i k_i s_i^2)^2}{\sum_i (k_i s_i^2)^2 / \nu_i}, \quad (2.17)$$

where s_i^2 are variance estimates, k_i are positive weights, and ν_i are their component degrees of freedom.

The final statistic uses a Student distribution rather than a normal distribution. Small effective degrees of freedom produce heavier tails, reflecting greater uncertainty in the estimated denominator. As the component variance estimates become stable, the effective degrees of freedom increase and the Student reference approaches the

normal reference.

For a single scalar restriction, normal and chi-squared Wald notation are equivalent. If T is asymptotically standard normal, then T^2 is asymptotically χ_1^2 . The signed form is retained here because the parameter is the scalar difference $I(P) - I(Q)$, its sign is meaningful, and the Welch refinement replaces the normal reference for T by a Student reference. The $(r - 1)(c - 1)$ -degree chi-squared law used for one-table independence is a different second-order result.

2.5 Prior comparisons of entropy and mutual information

Hutcheson applied a Welch–Satterthwaite architecture to differences between Shannon diversity values (Hutcheson, 1970). The method combines an entropy difference, analytic variance estimates, and a Student reference. It is the closest direct predecessor to the present study because entropy and MI are both nonlinear functionals of categorical probabilities.

The analogy is useful but incomplete. Entropy depends directly on one probability vector. Discrete MI depends on the joint probabilities and on the two margins calculated from them. Moving probability into one cell therefore changes not only that cell’s pointwise MI but also pointwise values throughout its row and column. An MI-specific variance calculation must retain these indirect effects.

The comparison of MI-related quantities is also established. Mora and Ruiz-Castillo studied the estimation and comparison of an MI-based segregation index (Mora and Ruiz-Castillo, 2009, 2011). Their work places MI comparison within a broader literature on nonlinear inequality and segregation measures. Related research combines analytic bias adjustments, asymptotic tests, and bootstrap refinements for such indices (Allen et al., 2015).

In neuroscience, MI estimates have been compared using jackknife uncertainty estimates (Hart and Giszter, 2010). Other applications apply ordinary mean tests to a collection of replicated MI estimates rather than deriving uncertainty from a single multinomial table in each population. These uses demonstrate the practical demand for differential MI inference, but they do not provide the variance-functional degrees of freedom developed here.

Examples of the replicated-estimate approach occur across several fields. MI values from randomised astronomical samples have been compared with a Student test (Sarkar and Pandey, 2020), and per-neuron information measurements have

been analysed with Welch comparisons (Prince et al., 2021). Partitioned continuous-data MI estimates have also been compared using unequal-variance mean tests (Martin et al., 2026). In each case, replication supplies a sample of MI values. The present problem instead begins with one multinomial table from each population and must derive uncertainty from the cell counts themselves.

Information-theoretic statistics and mean-based tests have also been used as parallel detectors rather than as parts of one inferential construction (Mather et al., 2013). These applications show that the combination of MI and Welch terminology is not a novelty claim. The narrower question is whether the variance-component logic can be derived for the plug-in MI functional itself.

Hutcheson’s result also defines an important novelty boundary. Substituting MI for entropy in a Welch-like template is not by itself a new statistical principle. MI introduces additional structure because its pointwise values depend on both joint and marginal probabilities. The sampling uncertainty of the resulting MI variance estimate must account for those changing margins.

2.6 Resampling and the weak null

Bootstrap methods can approximate the sampling distribution of nonlinear indices by repeatedly sampling from fitted empirical populations. They are flexible but computationally more expensive than an analytic table scan. Their accuracy also depends on how well the empirical distribution represents rare cells and unknown support.

The bootstrap and permutation address different sources of uncertainty. A multinomial bootstrap samples each fitted table separately and approximates the estimator’s repeated-sampling distribution. A label permutation pools the observations and reallocates them between groups. The latter construction is exact only when the pooled observations are exchangeable under the null. Bootstrap refinements have improved inference for nonlinear segregation and entropy measures, which makes them valuable future comparators (Allen et al., 2015; Palma et al., 2022). They do not remove the need to state which null is being approximated.

Permutation requires a separate distinction. Pooling two samples and permuting their group labels is exact under the strong null $P = Q$, because the observations are exchangeable. Under the weak null $I(P) = I(Q)$, the two complete distributions may differ, so raw label permutation generally targets the wrong null. Studentized permutation can recover asymptotic validity for some weak-null parameter comparisons (Chung and Romano, 2013), but it is not the standard one-table MI shuffle

used to test independence.

The primary thesis comparison therefore uses two transparent analytic baselines that share the same effect estimate and standard error as the proposed method. Resampling remains relevant for future confirmatory work, but it is not treated as an exact finite-sample oracle for the unrestricted weak null.

2.7 Literature synthesis and methodological gap

The literature establishes all of the broad ingredients used in this thesis: plug-in MI estimation and bias correction, first-order influence-function variance, Wald comparison of smooth functionals, and Welch–Satterthwaite moment matching. It also establishes prior comparisons of entropy and MI-related indices.

No single ingredient is presented as new. Shannon established MI; Miller and later authors developed finite-sample bias analysis; influence-function and delta-method theory provide first-order variance; and Welch, Satterthwaite, and Hutcheson provide the variance-component architecture. Prior MI comparison work establishes the scientific problem. The thesis contribution is the particular MI variance-functional derivation needed to connect these pieces, followed by a controlled evaluation under distinct equal-MI populations.

The remaining gap is narrower. The ordinary Welch substitution assigns a classical sample-variance degree of freedom, such as $n-1$, to each variance component. The estimated MI variance is not an ordinary sample variance of fixed observations. It is a nonlinear function of the same empirical joint table used to calculate MI. A change in one observation modifies the cell probabilities, margins, pointwise-MI values, and their weighted variance.

The expanded method addresses this gap by differentiating the complete MI variance functional. The resulting observation-level sensitivity determines the sampling variance of the variance estimate and, through Satterthwaite moment matching, an MI-specific component degree of freedom. Chapters 3 and 4 develop this construction.

Chapter 3

Statistical Problem and Analytic Baselines

This chapter fixes the estimand, effect estimate, standard error, and two analytic baselines. All three methods evaluated in the thesis use the same bias-corrected MI difference and the same first-order standard error. They differ only in the reference distribution used to convert the standardized statistic into a p-value and confidence interval.

3.1 Data, estimand, and assumptions

Let

$$N^P = (N_{ij}^P), \quad N^Q = (N_{ij}^Q) \quad (3.1)$$

be independent $r \times c$ count tables with totals n_P and n_Q . The row and column categories have the same definitions in both populations. The empirical probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \quad (3.2)$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \quad (3.3)$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \quad (3.4)$$

The first-order construction uses the following assumptions.

Assumption 1 (Sampling). *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets). *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive support). *Every population cell has positive probability. Empirical zero cells are allowed, but widespread observed support loss is recorded as a diagnostic.*

Assumption 4 (Regular MI). *The first-order influence variances $V(P)$ and $V(Q)$ are positive. This excludes exact independence and other degenerate cases from the regular Student approximation.*

All logarithms are natural. MI differences and standard errors are therefore reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardized statistic unchanged.

3.2 Plug-in and bias-corrected MI

For the first table, define empirical pointwise MI by

$$\hat{\ell}_P(i, j) = \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right) \quad \text{when } \hat{p}_{ij} > 0. \quad (3.5)$$

The plug-in MI estimate is

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \hat{\ell}_P(i, j). \quad (3.6)$$

The same calculation gives $\hat{I}(Q)$.

Let

$$d = (r - 1)(c - 1). \quad (3.7)$$

Under fixed positive support, the leading bias correction gives

$$\hat{I}_{\text{BC}}(P) = \hat{I}(P) - \frac{d}{2n_P}, \quad \hat{I}_{\text{BC}}(Q) = \hat{I}(Q) - \frac{d}{2n_Q}. \quad (3.8)$$

The estimated effect is

$$\hat{\Delta}_{\text{BC}} = \hat{I}_{\text{BC}}(P) - \hat{I}_{\text{BC}}(Q). \quad (3.9)$$

The correction uses the declared alphabet dimensions rather than the observed nonempty dimensions. It may produce a negative corrected MI estimate in a sparse

sample. The implementation does not clip that estimate at zero because clipping would alter both the estimand and the sampling distribution.

3.3 First-order variance of plug-in MI

For population P , define

$$V(P) = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (3.10)$$

This is the population variance of pointwise MI. It governs the sampling variance of the complete plug-in estimator because the first-order MI error is an average of centred pointwise-MI contributions.

To see this directly, write one observation as $Z_a^{(P)} = (X_a^{(P)}, Y_a^{(P)})$. A first-order expansion gives

$$\hat{I}(P) - I(P) = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\ell_P(Z_a^{(P)}) - I(P)\} + o_p(n_P^{-1/2}). \quad (3.11)$$

The summands are independent, have mean zero, and have variance $V(P)$. Therefore,

$$\text{Var}\{\hat{I}(P)\} \approx \text{Var}\left[\frac{1}{n_P} \sum_{a=1}^{n_P} \{\ell_P(Z_a^{(P)}) - I(P)\}\right] \quad (3.12)$$

$$= \frac{1}{n_P^2} \sum_{a=1}^{n_P} V(P) \quad (3.13)$$

$$= \frac{V(P)}{n_P}. \quad (3.14)$$

Subtracting the fixed leading bias term in Equation (3.8) does not change this first-order variance.

The observed table estimates $V(P)$ by

$$\hat{V}(P) = \sum_{i,j: \hat{p}_{ij} > 0} \hat{p}_{ij} \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2. \quad (3.15)$$

Define $\hat{V}(Q)$ analogously.

3.4 Shared standard error and test statistic

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\text{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}. \quad (3.16)$$

The common standardized statistic is

$$T = \frac{\widehat{\Delta}_{\text{BC}}}{\widehat{\text{SE}}}. \quad (3.17)$$

Equations (3.9)–(3.17) are shared by normal Wald, simple Welch, and expanded Welch. This controlled comparison isolates the effect of the reference distribution.

3.5 Baseline 1: normal Wald

Under the regular first-order asymptotics,

$$T \xrightarrow{d} N(0, 1) \quad \text{under } H_0. \quad (3.18)$$

The two-sided normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \quad (3.19)$$

where Φ is the standard normal distribution function. A $100(1 - \alpha)\%$ confidence interval for Δ is

$$\widehat{\Delta}_{\text{BC}} \pm z_{1-\alpha/2} \widehat{\text{SE}}. \quad (3.20)$$

The normal reference is the large-sample limit shared by all three analytic methods. It does not assign a separate finite-sample uncertainty to $\widehat{V}(P)$ or $\widehat{V}(Q)$.

3.6 Baseline 2: simple Welch–Satterthwaite

Simple Welch treats each empirical MI influence variance as though it had the same component degrees of freedom as an ordinary sample variance:

$$\nu_P^{\text{simple}} = n_P - 1, \quad \nu_Q^{\text{simple}} = n_Q - 1. \quad (3.21)$$

Substitution into the Welch–Satterthwaite equation gives

$$\hat{\nu}_{\text{simple}} = \frac{\{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q\}^2}{\{\hat{V}(P)/n_P\}^2/(n_P - 1) + \{\hat{V}(Q)/n_Q\}^2/(n_Q - 1)}. \quad (3.22)$$

The two-sided p-value is

$$p_{\text{simple}} = 2\{1 - F_{t_{\hat{\nu}_{\text{simple}}}}(|T|)\}. \quad (3.23)$$

The Student reference has heavier tails than the normal reference, so simple Welch is weakly more conservative at a fixed observed $|T|$. Its limitation is the component degrees of freedom. The value $n - 1$ has a classical basis for an ordinary sample variance under normal sampling. The quantity $\hat{V}(P)$ is instead a nonlinear plug-in functional whose pointwise scores change with the empirical table.

3.7 What the expanded method must calculate

The shared statistic already supplies the estimated effect and its first-order standard error. The remaining task is to replace the generic values $n_P - 1$ and $n_Q - 1$ by component degrees of freedom that describe the sampling reliability of $\hat{V}(P)$ and $\hat{V}(Q)$.

Satterthwaite moment matching requires the first two moments of each positive variance estimate. To first order,

$$\mathbb{E}\{\hat{V}(P)\} \approx V(P). \quad (3.24)$$

Chapter 4 derives $\text{Var}\{\hat{V}(P)\}$ by calculating how the complete variance functional changes when one observation changes the joint table.

Chapter 4

Expanded Welch–Satterthwaite Method

The expanded method retains the bias-corrected difference, standard error, and test statistic from Chapter 3. It changes only the component degrees of freedom used in the Welch–Satterthwaite reference. This chapter derives those degrees of freedom from the sampling variability of the complete MI variance estimator.

4.1 Roadmap from the Welch–Satterthwaite equation

For independent variance estimates s_i^2 , positive weights k_i , and component degrees of freedom ν_i , the Welch–Satterthwaite equation is

$$\nu = \frac{(\sum_i k_i s_i^2)^2}{\sum_i (k_i s_i^2)^2 / \nu_i}. \quad (4.1)$$

For the two MI estimates, the variance contributions are

$$k_P s_P^2 = \frac{\hat{V}(P)}{n_P}, \quad k_Q s_Q^2 = \frac{\hat{V}(Q)}{n_Q}. \quad (4.2)$$

These are the two terms in the squared standard error. The remaining quantities are the component degrees of freedom ν_P and ν_Q , which should measure how reliably $\hat{V}(P)$ and $\hat{V}(Q)$ estimate their population counterparts.

The derivation proceeds in four stages. First, Satterthwaite moment matching expresses ν_P in terms of the mean and variance of $\hat{V}(P)$. Second, a one-cell perturbation gives the observation-level sensitivity of $V(P)$. Third, the variability

of these sensitivities gives the sampling variance of $\hat{V}(P)$. Finally, the two estimated component degrees of freedom are combined through Equation (4.1).

4.2 Component degrees of freedom by moment matching

Satterthwaite represents a positive variance estimate by a scaled chi-squared variable. For the first population, write the working model as

$$\hat{V}(P) \sim E\{\hat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}. \quad (4.3)$$

The model is a moment-matching approximation. It does not assert that the finite-sample distribution of $\hat{V}(P)$ is exactly chi-squared.

For $X \sim \chi_{\nu_P}^2$,

$$E(X) = \nu_P, \quad \text{Var}(X) = 2\nu_P. \quad (4.4)$$

The scaling in Equation (4.3) therefore matches the mean automatically:

$$E\left[E\{\hat{V}(P)\} \frac{X}{\nu_P}\right] = E\{\hat{V}(P)\}. \quad (4.5)$$

Its variance is

$$\text{Var}\left[E\{\hat{V}(P)\} \frac{X}{\nu_P}\right] = \frac{[E\{\hat{V}(P)\}]^2}{\nu_P^2} \text{Var}(X) \quad (4.6)$$

$$= \frac{2[E\{\hat{V}(P)\}]^2}{\nu_P}. \quad (4.7)$$

Matching this quantity to $\text{Var}\{\hat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[E\{\hat{V}(P)\}]^2}{\text{Var}\{\hat{V}(P)\}}}. \quad (4.8)$$

To first order, $E\{\hat{V}(P)\} \approx V(P)$. The required calculation is therefore the sampling variance of $\hat{V}(P)$.

4.3 A one-cell perturbation of the distribution

The sampling behaviour of a smooth plug-in functional can be obtained by measuring its first-order response to one observation. Fix a cell (x, y) and define the perturbed

distribution

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(x,y)}, \quad (4.9)$$

where $\delta_{(x,y)}$ places probability one on cell (x, y) . At $\varepsilon = 0$, the distribution is P . Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j) ,

$$p_{ij}(\varepsilon) = (1 - \varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \quad (4.10)$$

The joint and marginal probability derivatives at the original distribution are

$$\left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \quad (4.11)$$

$$\left. \frac{dp_{i+}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \quad (4.12)$$

$$\left. \frac{dp_{+j}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \quad (4.13)$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \quad (4.14)$$

4.4 Change in pointwise mutual information and MI

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \quad (4.15)$$

Differentiating the three logarithms and substituting the probability derivatives gives

$$\left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j) \right|_0 = \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1. \quad (4.16)$$

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j). \quad (4.17)$$

The product rule separates its derivative into the change in probability weights and

the change in pointwise-MI values:

$$\begin{aligned} \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \{ \mathbf{1}(i=x, j=y) - p_{ij} \} \ell_P(i, j) \\ &\quad + \sum_{i,j} p_{ij} \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.18)$$

The first sum is $\ell_P(x, y) - I(P)$. Substitution of Equation (4.16) into the second sum gives $1 - 1 - 1 + 1 = 0$. Hence

$$\boxed{\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P)}. \quad (4.19)$$

This result recovers the centred pointwise-MI influence function used in Chapter 3.

4.5 Change in the MI variance

The target is the first-order change in

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\}. \quad (4.20)$$

Write this variance as

$$V(P) = M_2(P) - I(P)^2, \quad M_2(P) = \text{E}_P\{\ell_P(X, Y)^2\}. \quad (4.21)$$

Define the observation-level variance sensitivity by

$$g_P(x, y) = \left. \frac{dV(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.22)$$

Differentiating Equation (4.21) gives

$$g_P(x, y) = \left. \frac{dM_2(P_\varepsilon)}{d\varepsilon} \right|_0 - 2I(P) \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.23)$$

It remains to differentiate

$$M_2(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j)^2. \quad (4.24)$$

The product rule gives

$$\begin{aligned} \left. \frac{dM_2(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 \ell_P(i,j)^2 \\ &\quad + 2 \sum_{i,j} p_{ij} \ell_P(i,j) \left. \frac{d\ell_{P_\varepsilon}(i,j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.25)$$

The first sum is

$$\ell_P(x,y)^2 - M_2(P). \quad (4.26)$$

For the second sum, define the probability-weighted row and column means

$$E_P\{\ell_P(X,Y) \mid X = x\} = \frac{\sum_j p_{xj} \ell_P(x,j)}{p_{x+}}, \quad (4.27)$$

$$E_P\{\ell_P(X,Y) \mid Y = y\} = \frac{\sum_i p_{iy} \ell_P(i,y)}{p_{+y}}. \quad (4.28)$$

Substitution of Equation (4.16) gives

$$\begin{aligned} \sum_{i,j} p_{ij} \ell_P(i,j) \left. \frac{d\ell_{P_\varepsilon}(i,j)}{d\varepsilon} \right|_0 \\ = \ell_P(x,y) - E_P\{\ell_P(X,Y) \mid X = x\} - E_P\{\ell_P(X,Y) \mid Y = y\} + I(P). \end{aligned} \quad (4.29)$$

Combining Equations (4.19), (4.23), (4.26), and (4.29) yields

$$\begin{aligned} g_P(x,y) &= \{\ell_P(x,y) - I(P)\}^2 - V(P) \\ &\quad + 2 \left[\ell_P(x,y) - E_P\{\ell_P(X,Y) \mid X = x\} \right. \\ &\quad \left. - E_P\{\ell_P(X,Y) \mid Y = y\} + I(P) \right]. \end{aligned} \quad (4.30)$$

The first line is the change that would arise from the centred squared pointwise-MI value. The remaining terms account for the row and column margins that change with the selected cell.

4.6 Sampling variance of the MI variance estimator

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \text{Var}_P\{g_P(X,Y)\}. \quad (4.31)$$

The mean of an influence function is zero. Centring explicitly gives the same variance if numerical substitution produces a small nonzero mean.

A first-order expansion of the variance estimator is

$$\hat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \quad (4.32)$$

The observations are independent, so

$$\text{Var}\{\hat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \text{Var}_P\{g_P(Z_a^{(P)})\} \quad (4.33)$$

$$= \frac{\tau^2(P)}{n_P}. \quad (4.34)$$

Substitution into Equation (4.8) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \quad (4.35)$$

When $\hat{V}(P)$ is stable between samples, $\tau^2(P)$ is small and the component degrees of freedom are large. An unstable variance estimate produces smaller degrees of freedom and a heavier-tailed final reference.

4.7 Table-based component degrees of freedom

The observed table replaces every population quantity in Equation (4.30) by its empirical counterpart. For each occupied cell,

$$\begin{aligned} \hat{g}_P(i, j) &= \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \\ &+ 2 \left[\hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right]. \end{aligned} \quad (4.36)$$

The empirical mean and variance of these cell sensitivities are

$$\bar{g}_P = \sum_{i,j} \hat{p}_{ij} \hat{g}_P(i, j), \quad (4.37)$$

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \{\hat{g}_P(i, j) - \bar{g}_P\}^2. \quad (4.38)$$

The estimated component degree of freedom is

$$\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}. \quad (4.39)$$

Repeating the same calculation for the second table gives $\hat{\nu}_Q$.

4.8 Combined degrees of freedom and final test

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\hat{\nu}_{\text{expanded}} = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\left\{ \hat{V}(P)/n_P \right\}^2 / \hat{\nu}_P + \left\{ \hat{V}(Q)/n_Q \right\}^2 / \hat{\nu}_Q}. \quad (4.40)$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2 \left\{ 1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right\}, \quad (4.41)$$

and the corresponding confidence interval is

$$\hat{\Delta}_{\text{BC}} \pm t_{\hat{\nu}_{\text{expanded}}, 1-\alpha/2} \widehat{\text{SE}}. \quad (4.42)$$

The expanded calculation does not change $\hat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$, or T . It changes only the quantile used to evaluate that statistic.

4.9 Worked count-table example

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \quad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \quad (4.43)$$

Their totals are $n_P = 105$ and $n_Q = 100$. The plug-in MI estimates are 0.13693 and 0.25848 nats. With $d = (3 - 1)(3 - 1) = 4$, the corrected difference is

$$\hat{\Delta}_{\text{BC}} = \left(0.13693 - \frac{4}{2(105)} \right) - \left(0.25848 - \frac{4}{2(100)} \right) = -0.12059. \quad (4.44)$$

The empirical influence variances are $\hat{V}(P) = 0.27189$ and $\hat{V}(Q) = 0.43371$. Hence

$$\widehat{\text{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100} \right)^{1/2} = 0.08323, \quad T = -1.44898. \quad (4.45)$$

Normal Wald gives $p = 0.14734$. Simple Welch uses total effective degrees of freedom 188.53 and gives $p = 0.14900$.

For expanded Welch, the table-based sensitivity calculation gives component degrees of freedom

$$\hat{\nu}_P = 16.65, \quad \hat{\nu}_Q = 45.86. \quad (4.46)$$

Equation (4.40) combines them into $\hat{\nu}_{\text{expanded}} = 59.03$, giving $p = 0.15264$. The three methods reach the same substantive conclusion for these tables, but the example shows where the expanded reference differs: it regards the two MI variance estimates as less stable than the generic $n - 1$ assignment does.

4.10 Operational algorithm and complexity

Table 4.1 summarises one complete comparison.

Table 4.1: Operational expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned independent count tables and their declared alphabet dimensions.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\hat{V}(P)$, $\hat{V}(Q)$, the shared standard error, and T .
4	Compute the row and column pointwise-MI means needed in $\hat{g}_P$ and $\hat{g}_Q$.
5	Compute $\hat{\tau}^2(P)$, $\hat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.40).
7	Return the Student p-value, confidence interval, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$. The method is deterministic once the two count tables are given.

4.11 Theoretical status

The influence-function calculations in Equations (4.19) and (4.30) are directional derivatives of the corresponding population functionals. Under fixed positive support, they provide the usual first-order asymptotic linear representations.

The scaled chi-squared model in Equation (4.3) and the final Student reference are working approximations. They match the first two moments of the estimated variance rather than providing an exact finite-sample law. Under regular large-sample conditions, $\hat{\nu}_P$, $\hat{\nu}_Q$, and $\hat{\nu}_{\text{expanded}}$ increase, so the Student reference approaches the same normal limit as Wald inference.

The method can fail when the empirical table loses too much support, when the estimated MI variance is zero, or when the estimated sensitivity variance is not positive and finite. These cases are recorded as invalid calculations rather than silently replaced by a different test. The validation in Chapters 5 and 6 evaluates how often these conditions occur and whether the finite-sample correction improves calibration.

Chapter 5

Experimental Design

The derivation in Chapter 4 establishes a plausible finite-sample reference but not an exact distributional theorem. Its value must therefore be assessed through repeated sampling. This chapter describes one controlled experiment designed to answer a direct question: when two different categorical populations have exactly equal positive MI, how often does each analytic method incorrectly reject equality?

5.1 Validation principle

For each scenario, P and Q are fixed $r \times c$ population probability tables satisfying

$$I(P) = I(Q). \quad (5.1)$$

Independent count tables are repeatedly drawn as

$$N^P \sim \text{Multinomial}(n_P, P), \quad N^Q \sim \text{Multinomial}(n_Q, Q). \quad (5.2)$$

Normal Wald, simple Welch, and expanded Welch analyse the same sampled table pair in every replicate. This pairing removes Monte Carlo variation from between-method differences and isolates the effect of the reference distribution.

The primary significance level is $\alpha = 0.05$. A calibrated method should reject Equation (5.1) in approximately 5% of repeated samples. Additional checks use $\alpha = 0.10$, $\alpha = 0.01$, and a full rejection-calibration curve from 0 to 0.10.

5.2 Constructing distinct populations with equal MI

The weak null permits $P \neq Q$. Each scenario therefore requires two different joint distributions with numerically equal MI. The construction separates marginal shape from association strength.

5.2.1 Random margins

For a variable with m categories, a symmetric Dirichlet distribution generates a positive probability vector

$$(u_1, \dots, u_m) \sim \text{Dirichlet}(a, \dots, a). \quad (5.3)$$

The concentration a controls how similar the category probabilities are. Large values produce margins close to uniform. Values below one commonly produce several rare categories. The implementation rejects margins with a probability below 10^{-4} or above 0.98 to avoid numerical degeneracy while retaining substantial skewness.

The row and column margins for P and Q are generated independently. The two populations can therefore have different marginal distributions even when their MI values are equal.

5.2.2 Random association pattern

Margins alone define the independent table. To introduce association, the generator draws a matrix $A = (A_{ij})$ of independent standard-normal values, removes its row and column means, and standardises its root-mean-square magnitude. This produces a non-additive interaction pattern that cannot be absorbed into the margins.

For association strength λ , define a positive log-linear kernel

$$K_{ij}(\lambda) = \exp(\lambda A_{ij}). \quad (5.4)$$

Iterative proportional fitting finds positive scaling vectors u and v such that

$$p_{ij}(\lambda) = u_i K_{ij}(\lambda) v_j \quad (5.5)$$

has the required row and column margins. A one-dimensional root finder then

selects λ so that

$$I\{P(\lambda)\} = I_{\text{target}}. \quad (5.6)$$

The same procedure is run independently for Q . Its margins, interaction pattern, and fitted association strength differ from those of P , but its target MI is identical.

A generated pair is retained only when all cells have positive population probability, both first-order MI variances are positive, and

$$\|P - Q\|_1 = \sum_{i,j} |p_{ij} - q_{ij}| \geq 0.05. \quad (5.7)$$

The largest absolute difference between $I(P)$ and $I(Q)$ among the 60 accepted pairs was 8.5×10^{-14} nats. Numerical mismatch in the null was therefore negligible relative to sampling variation.

5.3 Table shapes and regimes

Every regime uses six table shapes:

$$2 \times 2, \quad 2 \times 5, \quad 3 \times 3, \quad 3 \times 5, \quad 5 \times 5, \quad 8 \times 8. \quad (5.8)$$

Each regime has two variants. The complete design contains

$$6 \text{ shapes} \times 5 \text{ regimes} \times 2 \text{ variants} = 60 \quad (5.9)$$

fixed equal-MI population pairs. Both sample sizes lie between 50 and 1,000. The two target values, 0.10 and 0.15 nats, keep the primary experiment away from the nonregular independence boundary.

Table 5.1 gives the exact construction rules. The concentrations (a_P, a_Q) apply to both row and column margins in the corresponding populations. The ratio is $n_Q : n_P$.

For the well-sampled and moderate regimes, sample size is fixed from the shape and design constants:

$$n_P = \max(n_{\min}, drc), \quad n_Q = \rho n_P, \quad (5.10)$$

where d is the planned average number of observations per cell and $\rho = n_Q/n_P$. This rule is chosen before the population tables are drawn.

For the three sparse regimes, sample sizes are selected after the population

Table 5.1: Exact equal-MI regime specifications.

Regime and variant	MI	(a_P, a_Q)	Ratio	Sampling rule
Well sampled 1	0.10	(50, 50)	1 : 1	15 observations per cell; $n_P \geq 200$
Well sampled 2	0.15	(50, 50)	1 : 1	15 observations per cell; $n_P \geq 200$
Moderate 1	0.10	(8, 4)	1 : 1	8 observations per cell; $n_P \geq 100$
Moderate 2	0.15	(8, 2)	2 : 1	6 observations per cell; $n_P \geq 100$
Highly skewed and sparse 1	0.10	(2, 2)	1 : 1	Both minima in [1, 5)
Highly skewed and sparse 2	0.15	(10, 2)	2 : 1	Both minima in [1, 5)
Ultra-skewed and sparse 1	0.10	(0.8, 0.8)	1 : 1	Both minima in [0.05, 1)
Ultra-skewed and sparse 2	0.15	(10, 0.8)	10 : 1	Both minima in [0.05, 1)
Widespread sparsity 1	0.10	(0.35, 0.35)	1 : 1	25–50% of cells below 1 and at least 50% below 5
Widespread sparsity 2	0.15	(1, 0.35)	2 : 1	Same cell-wide sparsity constraints

tables are constructed. The selection uses true joint expected counts

$$E_{ij}^P = n_P p_{ij}, \quad E_{ij}^Q = n_Q q_{ij} \quad (5.11)$$

to meet the stated rare-cell or cell-wide sparsity constraint. The resulting sample sizes and smallest expected counts are summarised in Table 5.2.

Table 5.2: Realised sample-size and expected-count ranges across the six shapes.

Regime and variant	n_P	n_Q	Smallest expected count
Well sampled 1	200–960	200–960	2.140–18.228
Well sampled 2	200–960	200–960	1.385–15.015
Moderate 1	100–512	100–512	0.790–6.927
Moderate 2	100–384	200–768	0.057–1.178
Highly skewed and sparse 1	52–975	52–975	1.025–4.072
Highly skewed and sparse 2	54–460	108–920	1.096–4.201
Ultra-skewed and sparse 1	64–604	64–604	0.052–0.413
Ultra-skewed and sparse 2	51–75	510–750	0.057–0.632
Widespread sparsity 1	50–435	50–435	below 0.001–0.042
Widespread sparsity 2	50–414	100–828	below 0.001–0.312

Widespread sparsity is a boundary check rather than a recommended operating regime. Its purpose is to reveal what happens when many observed cells, and

occasionally complete empirical margins, can be empty.

5.4 Replication and evaluation metrics

Each fixed population pair receives 10,000 independently sampled table pairs:

$$60 \times 10,000 = 600,000 \quad (5.12)$$

null replicates, containing 1.2 million individual count tables. The population-generation seed is 2,026,080,501 and the sampling seed is 2,026,080,502. Scenario-specific seeds are derived deterministically from these values.

At a true rejection probability of 0.05, 10,000 replicates give Monte Carlo standard error

$$\sqrt{\frac{0.05(1 - 0.05)}{10,000}} \approx 0.00218 \quad (5.13)$$

for one scenario. Wilson intervals are reported for scenario-level rejection rates.

The experiment uses the following metrics.

False-positive rate.

The fraction of valid null replicates with p-value no larger than the nominal significance level.

Mean absolute calibration error.

For a regime with scenario FPRs $\hat{f}_s$,

$$\text{MAE}(\alpha) = \frac{1}{S} \sum_{s=1}^S |\hat{f}_s - \alpha|. \quad (5.14)$$

This metric prevents liberal and conservative scenario errors from cancelling.

Rejection-calibration curve.

The empirical rejection rate is evaluated at nominal levels from 0 to 0.10 in increments of 0.001. A calibrated test follows the diagonal.

Coverage.

The fraction of valid null replicates whose method-specific 95% confidence interval contains the true difference, zero.

Valid rate.

The fraction of replicates producing a finite statistic and p-value and, where required, positive finite effective degrees of freedom.

Power. The fraction of valid alternative replicates that correctly reject equal MI at $\alpha = 0.05$.

Regime summaries weight every fixed population pair equally. Scenario-level results remain available so that an average improvement cannot conceal a specific failure.

5.5 Power design

The power experiment uses five 3×3 alternatives. Population P has uniform row and column margins and $I(P) = 0.05$ nats. Population Q has margins $(0.9, 0.05, 0.05)$ and a larger MI. Both use the same ordinal interaction pattern, which keeps the comparison focused on MI difference and sample size. Table 5.3 gives the exact settings; each receives 10,000 replicates.

Table 5.3: Alternative-hypothesis settings for the power experiment.

$I(P)$	$I(Q)$	n_P	n_Q	Minimum expected counts (P, Q)
0.05	0.07	300	300	(18.217, 1.543)
0.05	0.10	150	150	(9.108, 0.926)
0.05	0.10	300	300	(18.217, 1.853)
0.05	0.10	600	600	(36.433, 3.705)
0.05	0.15	300	300	(18.217, 1.143)

5.6 Runtime design

Runtime is measured after count-table construction. One fixed table pair is used for each of four shapes: 2×2 , 3×3 , 5×5 , and 8×8 . Each complete method call is warmed up and then timed 200 times. The median is reported. All methods use the same vectorised implementation path, so the comparison measures the additional reference calculation rather than differences between software packages.

5.7 Reproducibility controls

The full experiment records the fixed population tables, sample sizes, scenario seeds, simulation seeds, method outputs, software versions, and SHA-256 hash of the experiment script. Population construction and evaluation use separate

deterministic random-number streams. No empirical result from a replicate is used to alter its generating population or select its method.

The experiment used Python 3.13.5, NumPy 2.4.4, SciPy 1.17.1, and pandas 3.0.3. Appendix [E](#) gives the commands and output map.

Chapter 6

Results

This chapter evaluates the three reference distributions while holding the MI difference and standard error fixed. Rejection calibration is the primary outcome. Coverage, power, validity, and runtime describe the associated trade-offs.

6.1 Rejection calibration across regimes

Figure 6.1 plots actual rejection rate against the nominal significance level. The diagonal represents perfect calibration. A curve above the diagonal is liberal and rejects too often; a curve below it is conservative. Each method’s line is the equal-weight mean across the 12 fixed population pairs in that regime. Shading spans the 10th to 90th percentiles of the scenario-specific rates.

The three curves are close in the well-sampled control. The expanded reference moves the moderate and sparse curves toward the diagonal. The largest visible change occurs in the ultra-skewed and widespread-sparsity regimes, where the normal Wald and simple Welch curves are substantially liberal.

Table 6.1 separates mean FPR from mean absolute scenario error at $\alpha = 0.05$. Mean FPR describes direction, while mean absolute error measures scenario-level accuracy without cancellation.

Table 6.1: Calibration at $\alpha = 0.05$. Error change is the relative reduction in mean absolute error from normal Wald to expanded Welch; a negative value denotes deterioration.

Regime	Wald FPR	Simple FPR	Expanded FPR	Wald MAE	Expanded MAE	Change
Well sampled	0.05174	0.05112	0.04913	0.00317	0.00332	−4.5%
Moderate	0.06158	0.06025	0.05623	0.01158	0.00720	37.8%
Highly skewed and sparse	0.05547	0.05445	0.05198	0.00645	0.00456	29.3%
Ultra-skewed and sparse	0.07504	0.07208	0.06305	0.02504	0.01340	46.5%
Widespread sparsity	0.06621	0.06400	0.05687	0.02087	0.01277	38.8%

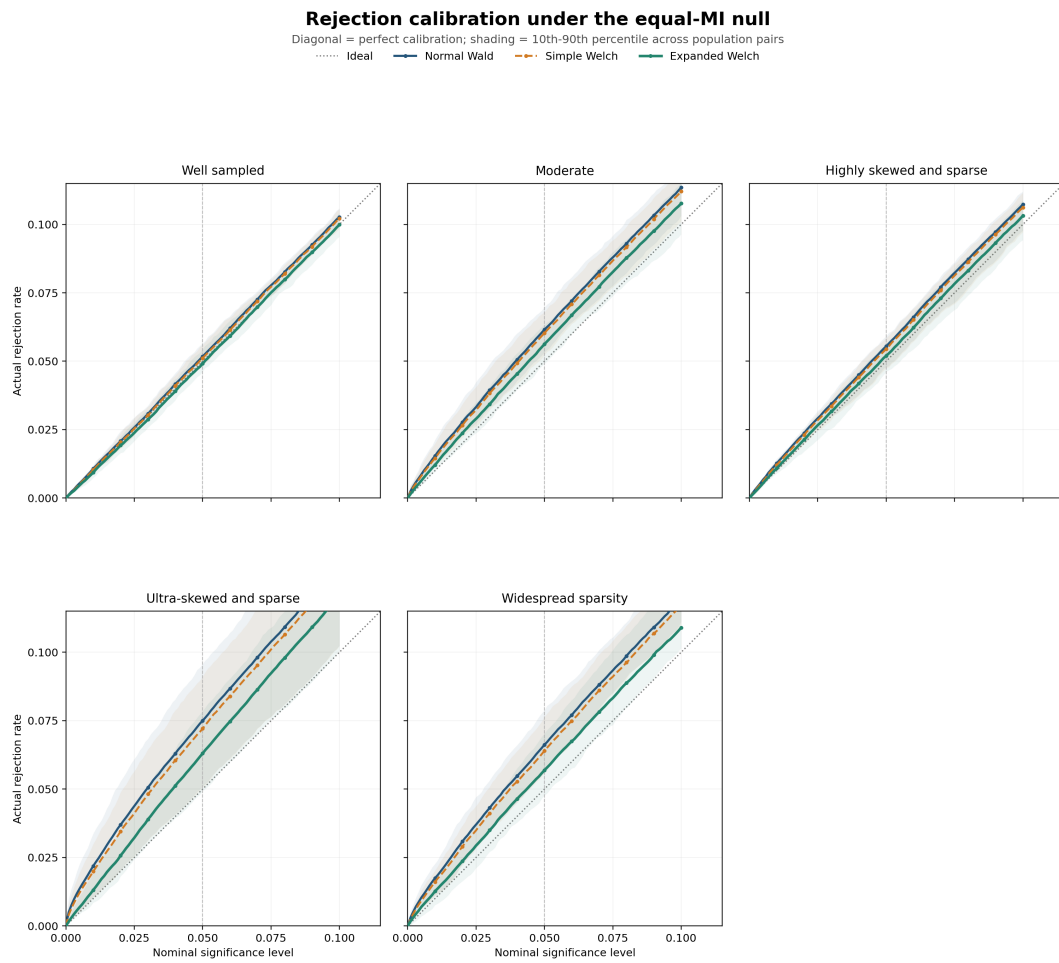


Figure 6.1: Rejection calibration under the equal-MI null. Each line is the mean across fixed population pairs in one regime, and each band spans the 10th to 90th percentiles across those pairs. The vertical guide marks $\alpha = 0.05$.

Normal Wald was already calibrated in the well-sampled control. Expanded Welch changed mean FPR from 0.05174 to 0.04913, but its mean absolute error was 4.5% larger because some scenarios became mildly conservative. The result is consistent with the intended asymptotic behaviour: when the component degrees of freedom are large, little correction is needed.

The moderate regime had normal Wald mean FPR 0.06158. Expanded Welch reduced it to 0.05623 and reduced mean absolute error by 37.8%. In the highly skewed and sparse regime, where both minimum expected counts lay between 1 and 5, expanded Welch reduced mean absolute error by 29.3%.

The largest relative improvement occurred in the ultra-skewed regime. Expanded Welch reduced mean FPR from 0.07504 to 0.06305 and mean absolute error by 46.5%. The method improved the result but did not restore exact calibration. The widespread-sparsity boundary showed a similar pattern: mean FPR fell from 0.06621 to 0.05687, while mean absolute error fell by 38.8%.

At $\alpha = 0.01$, the expanded method reduced mean absolute error by 54.2% in the moderate regime, 51.6% in the highly skewed and sparse regime, 66.3% in the ultra-skewed regime, and 48.5% under widespread sparsity. The heavier Student tails had a larger relative effect in the more extreme tail.

6.2 Variation within regimes

Regime averages can hide unequal performance between design variants. Table 6.2 reports mean FPR across the six table shapes for each fully specified variant.

Table 6.2: Mean FPR at $\alpha = 0.05$ by regime variant.

Variant	MI	Ratio	Wald	Simple	Expanded
Well sampled 1	0.10	1 : 1	0.04942	0.04882	0.04612
Well sampled 2	0.15	1 : 1	0.05407	0.05342	0.05215
Moderate 1	0.10	1 : 1	0.05772	0.05647	0.05222
Moderate 2	0.15	2 : 1	0.06545	0.06403	0.06025
Highly sparse 1	0.10	1 : 1	0.05185	0.05077	0.04797
Highly sparse 2	0.15	2 : 1	0.05908	0.05813	0.05598
Ultra-sparse 1	0.10	1 : 1	0.05775	0.05638	0.05267
Ultra-sparse 2	0.15	10 : 1	0.09233	0.08778	0.07343
Widespread sparsity 1	0.10	1 : 1	0.06190	0.05926	0.05057
Widespread sparsity 2	0.15	2 : 1	0.07053	0.06874	0.06317

Expanded Welch improved both variants of every difficult regime. The most severe remaining error was the 10 : 1 ultra-sparse variant. Its mean FPR fell from

0.09233 to 0.07343 but remained materially above 0.05. This scenario combines a very small first sample, rare cells, and extreme sample imbalance. The reference correction cannot remove all higher-order bias and support instability in that setting.

Figure 6.2 removes the averaging over shapes and compares absolute FPR error for every population pair. Points below the diagonal favour the comparator over normal Wald. Simple Welch improved 53 of the 60 scenarios, but most movements were small. Expanded Welch improved 48 scenarios and worsened 12; its improvements were larger in the difficult regimes, while six of the 12 well-sampled controls became slightly less accurate. This explains why expanded Welch has the best aggregate error even though it does not win the largest number of individual scenarios.

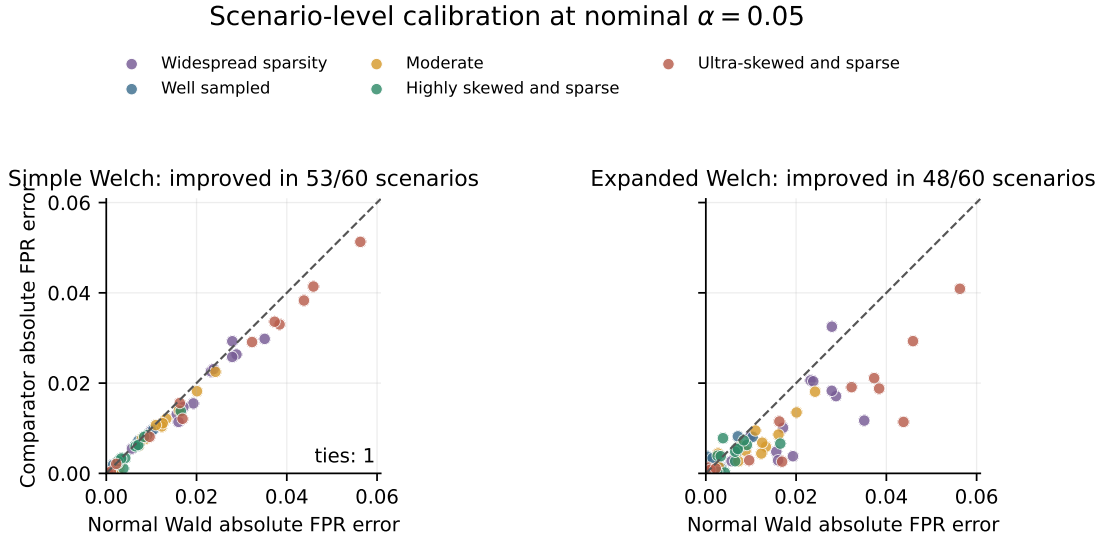


Figure 6.2: Scenario-level absolute FPR error at $\alpha = 0.05$. Each point is one fixed equal-MI population pair. Points below the diagonal have lower error than normal Wald; colour identifies the sampling regime.

6.3 Overall calibration, coverage, and validity

Table 6.3 aggregates all 60 fixed population pairs. Expanded Welch had the lowest mean absolute calibration error at each reported significance level and the coverage closest to 0.95.

All methods used the same base validity condition, which required positive finite first-order variance estimates. In the first four regimes the valid rate was effectively one. Under widespread sparsity it fell to 0.97447 because one sampled table could have zero estimated MI influence variance after severe support loss. Conditioning every method on the same valid table pairs keeps their reference comparison aligned,

Table 6.3: Overall results across the 60 equal-MI population pairs.

Method	MAE 0.10	MAE 0.05	MAE 0.01	Valid rate	Coverage
Normal Wald	0.01716	0.01342	0.00608	0.99489	0.93799
Simple Welch	0.01560	0.01194	0.00514	0.99489	0.93962
Expanded Welch	0.01179	0.00825	0.00278	0.99489	0.94455

but the reduced valid rate is itself a warning against relying on first-order inference in this regime.

Coverage followed the calibration result. Normal Wald’s overall 95% coverage was 0.93799, simple Welch increased it to 0.93962, and expanded Welch increased it to 0.94455. Expanded Welch remained slightly below nominal overall, which is consistent with the residual liberal behaviour in the most difficult variants.

6.4 Effective degrees of freedom

Figure 6.3 compares the median total degrees of freedom assigned by the two Student references in each scenario. All 60 points fall below the diagonal. The median simple-Welch value was 380.50, whereas the median expanded value was 71.56; the median ratio was 0.219. The expanded calculation therefore identifies substantially more uncertainty in the MI variance estimator than the generic $n - 1$ assignment.

Smaller degrees of freedom do not by themselves establish better inference. They describe the mechanism by which expanded Welch changes the reference tails. In the well-sampled scenarios, the median expanded value was still 117.03, for which the Student distribution is already close to normal. In the ultra-skewed scenarios it fell to 30.27, producing a more material correction.

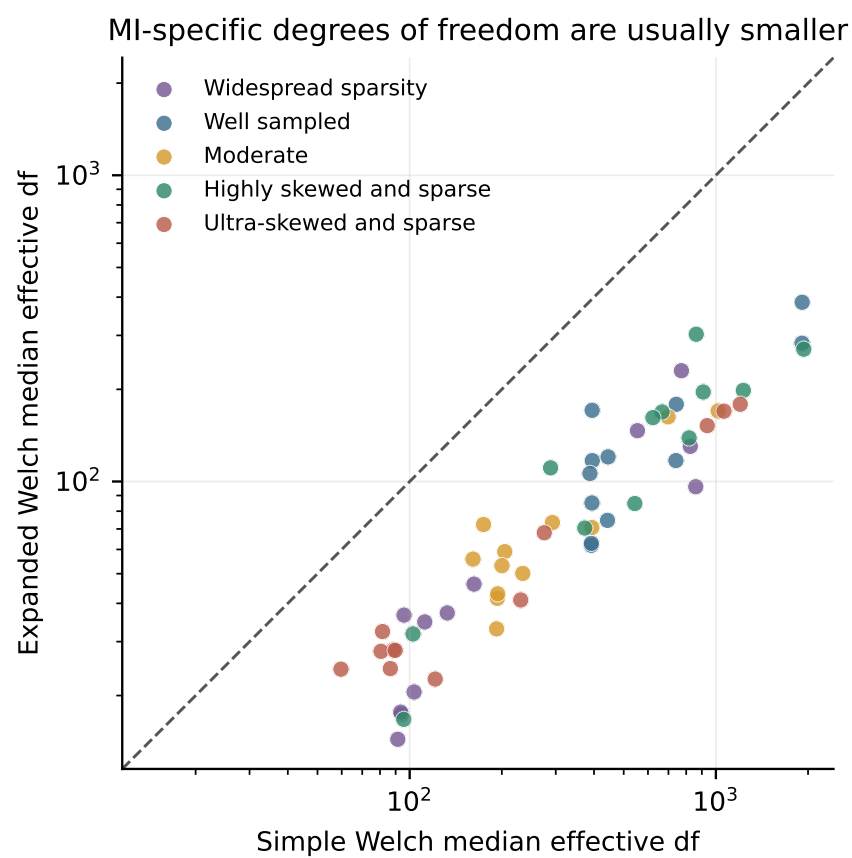


Figure 6.3: Median effective degrees of freedom for simple and expanded Welch in each fixed null scenario. Both axes are logarithmic; the dashed line denotes equality.

6.5 Power

Table 6.4 reports power at $\alpha = 0.05$. Power increased with both the true MI difference and sample size. Simple Welch was nearly indistinguishable from normal Wald. Expanded Welch had lower power because its heavier reference tails require a larger $|T|$ for rejection.

Figure 6.4 shows the same result graphically. Increasing the sample size from 150 to 600 at fixed $|\Delta| = 0.05$ raises power for all methods. The ordering between methods remains much smaller than the effect of sample size or the true MI difference.

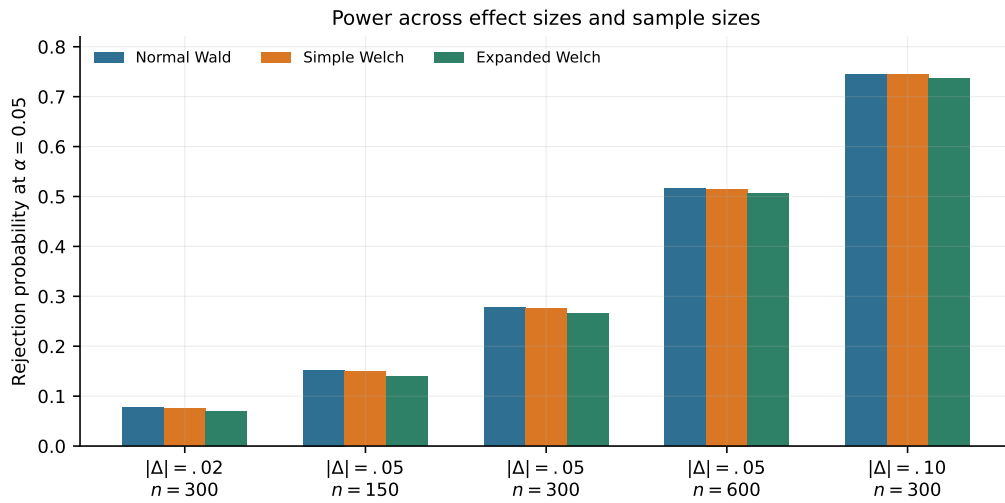


Figure 6.4: Power at $\alpha = 0.05$ for the five alternative scenarios. Here $n = n_P = n_Q$ and $|\Delta| = |I(P) - I(Q)|$.

Table 6.4: Power at $\alpha = 0.05$ for the five alternative scenarios.

$I(P)$	$I(Q)$	(n_P, n_Q)	Wald	Simple	Expanded
0.05	0.07	(300, 300)	0.0768	0.0759	0.0689
0.05	0.10	(150, 150)	0.1523	0.1498	0.1402
0.05	0.10	(300, 300)	0.2775	0.2761	0.2652
0.05	0.10	(600, 600)	0.5161	0.5151	0.5063
0.05	0.15	(300, 300)	0.7449	0.7437	0.7362

Relative to normal Wald, expanded Welch lost 1.02 percentage points of power on average and at most 1.23 percentage points in these scenarios. The power cost is small in absolute terms but is not zero. It is the expected trade-off for correcting a liberal reference with heavier tails.

6.6 Computational cost

Table 6.5 reports the median time for one complete method call after the count tables had been constructed. Expanded Welch required approximately 1.72–1.74 times the normal Wald runtime, but all measured times remained below 0.15 milliseconds.

Table 6.5: Median end-to-end method runtime over 200 repetitions.

Shape	(n_P, n_Q)	Wald (ms)	Simple (ms)	Expanded (ms)
2×2	(200, 200)	0.0842	0.0991	0.1451
3×3	(200, 200)	0.0844	0.0995	0.1456
5×5	(375, 375)	0.0837	0.0987	0.1452
8×8	(960, 960)	0.0858	0.1009	0.1481

The additional arithmetic is negligible in absolute terms. All methods are deterministic and have $O(rc)$ complexity. Runtime therefore differs in constant factors rather than asymptotic order.

6.7 Scaled chi-squared mechanism audit

The expanded derivation uses a scaled chi-squared model for the positive MI variance estimate. A separate experiment evaluated that working model on 128 population components from 64 fixed scenarios. For each component, 10,000 tables estimated finite-sample moments and an independent 10,000 tables evaluated distributional fit.

When every candidate distribution received the empirical mean and variance, the oracle scaled chi-squared model had mean Kolmogorov–Smirnov distance 0.0199, compared with 0.0196 for an oracle normal model and 0.0291 for an oracle lognormal model. Scaled chi-squared had lower upper-tail error than the normal model: 0.0045 versus 0.0054 at the 0.05 tail and 0.0026 versus 0.0038 at the 0.01 tail. It was therefore a plausible tail model but not uniformly the best whole-distribution family.

The direct population first-order model had mean KS distance 0.1622, showing that finite-sample error in the predicted moments is more important than the choice between chi-squared and normal shape families. The implemented plug-in component degrees of freedom were nevertheless close to empirical moment-matched values: their median ratio was 0.970. Appendix C reports the full diagnostic.

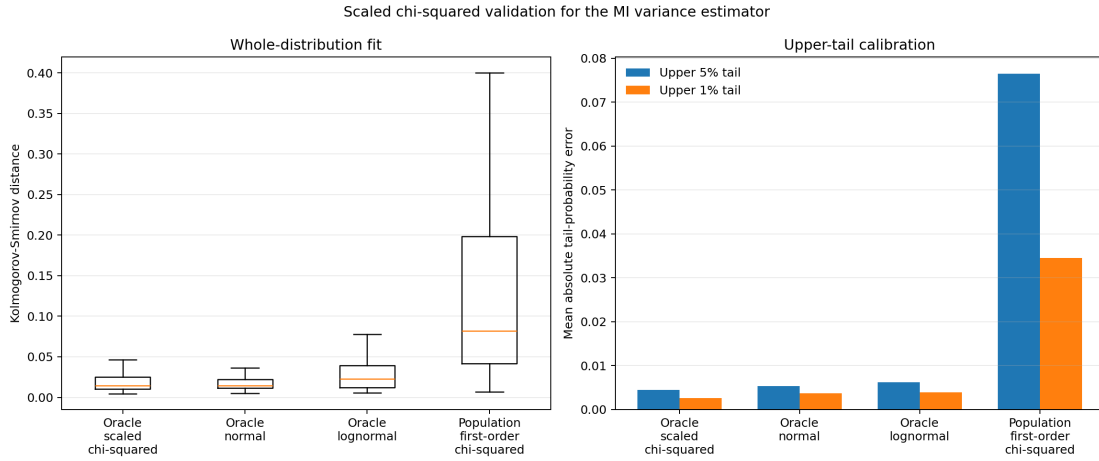


Figure 6.5: Audit of the scaled chi-squared working model. The left panel compares whole-distribution KS distance; the right panel compares absolute upper-tail probability error. Oracle models use empirical moments, whereas the population first-order model uses the derived moments.

6.8 Results summary

The empirical evidence gives a consistent but bounded result. Normal Wald was already accurate in well-sampled tables. Simple Welch made only a small change. Expanded Welch improved average calibration, coverage, and tail behaviour in the moderate and sparse regimes, with the largest gains under rare cells and unequal sample sizes. It incurred about one percentage point of average power loss and less than 0.07 milliseconds of additional median runtime. The most extreme sparse and imbalanced scenario remained liberal, and widespread support loss produced invalid first-order variance estimates in approximately 2.6% of table pairs.

Chapter 7

Discussion

This thesis asked whether the Welch–Satterthwaite principle can be adapted to compare mutual information between two independent categorical populations. The central problem is not estimation of the MI difference itself. It is calibration of the standardized difference when the two first-order variance contributions are estimated from finite, heterogeneous contingency tables. This chapter interprets the proposed solution and the evidence obtained for it.

7.1 Answers to the research questions

7.1.1 Method construction

Research Question 1 asked how to adapt Welch–Satterthwaite inference to the nonlinear plug-in MI variance estimator. The expanded method does this in two stages. For each population, it first measures the sampling variability of the estimated MI variance through the influence function g_P or g_Q . It then matches the mean and variance of that positive variance estimator to a scaled chi-squared variable. This produces the component degrees of freedom

$$\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}, \quad \hat{\nu}_Q = \frac{2n_Q \hat{V}(Q)^2}{\hat{\tau}^2(Q)}. \quad (7.1)$$

The ordinary Welch–Satterthwaite equation combines these values according to the two contributions to the squared standard error. The result is a Student reference whose tail weight is determined by the observed stability of the MI variance estimates. No resampling or numerical optimisation is required at test time.

The construction is MI-specific because g_P differentiates the complete variance functional. In particular, it includes the changes in pointwise MI caused by changing

the joint and marginal probabilities. Replacing this step by $n_P - 1$, as in a direct analogy with Welch’s test for means, does not represent the sampling uncertainty of the MI variance estimator.

7.1.2 Statistical performance

Research Question 2 asked whether expanded Welch improves type-I error and coverage. Across all 60 equal-MI population pairs, it reduced mean absolute false-positive-rate error at $\alpha = 0.05$ from 0.01342 for normal Wald and 0.01194 for simple Welch to 0.00825. Overall 95% coverage increased from 0.93799 for normal Wald to 0.94455.

The improvement was concentrated in the regimes for which the method was designed. Expanded Welch reduced calibration error by 29.3–46.5% relative to normal Wald in the highly skewed, ultra-skewed, and widespread-sparsity regimes. It also improved the moderate regime by 37.8%. In the well-sampled control, normal Wald was already accurate and expanded Welch was effectively similar, although its mean absolute error was 4.5% larger. The evidence therefore supports improved average calibration in difficult positive-MI settings, not uniform dominance for every population pair.

The most extreme scenario remained outside satisfactory calibration. Under ultra-sparse cells and a 10 : 1 sample-size ratio, expanded Welch reduced mean FPR from 0.09233 to 0.07343 but did not reach the nominal value of 0.05. The reference correction addresses uncertainty in the estimated variance; it does not remove all finite-sample bias, discreteness, or empirical support loss.

7.1.3 Practical trade-offs

Research Question 3 asked about power, runtime, validity, and operating range. The heavier Student tails reduced power by 1.02 percentage points on average across the five alternatives and by at most 1.23 points. This is a small but genuine cost. A liberal null reference can appear more powerful partly because it rejects too often under both the null and alternative. The calibration and power results must therefore be interpreted together.

Expanded Welch took approximately 0.15 milliseconds per table pair in the timed implementation, compared with 0.08 milliseconds for normal Wald. Both methods have $O(rc)$ time and memory complexity. The correction is therefore about 1.7 times slower in the tested implementation but remains negligible in absolute cost and avoids the multiplicative cost of resampling.

The valid rate was effectively one outside widespread sparsity. It fell to 0.97447 when support loss caused the empirical first-order variance to vanish in some replicates. A finite p-value is not the only criterion for a useful test; the underlying first-order variance must also be identifiable from the observed table.

7.2 Why the correction improves difficult cases

Normal Wald treats the estimated standard error as if it were known. This is a reasonable approximation when the table is well sampled, because repeated samples then produce similar empirical probabilities and similar variance estimates. Skewness, rare cells, and unequal sample sizes make the denominator of the test statistic less stable. A normal reference does not respond to that additional uncertainty.

Simple Welch adds finite degrees of freedom, but assigns them from sample sizes alone. Its reference changes little when two tables have the same sample sizes but very different cell probabilities. The primary results show the consequence: simple Welch generally lies between normal Wald and expanded Welch, but its improvement is modest in the most difficult regimes.

Expanded Welch estimates how strongly the complete MI variance functional changes when probability is moved towards each cell. Large variation among the resulting $g_P(i, j)$ values implies a volatile variance estimate, a smaller component degree of freedom, and a heavier-tailed Student reference. The correction therefore becomes strongest when the observed table indicates that the standard error itself is unreliable. As sampling improves, $\hat{\nu}_P$ and $\hat{\nu}_Q$ increase and the Student reference approaches the normal reference.

This mechanism also explains the residual failures. The derivation is first-order. When many cells disappear from the empirical support, both the point estimate and its estimated variance can be governed by higher-order and discrete effects. Changing only the reference tails cannot reconstruct information that is absent from the table.

7.3 Interpretation of the scaled chi-squared model

Satterthwaite's scaled chi-squared representation is a working model for the positive estimated variance, not an assertion that its exact finite-sample distribution is chi-squared. Its role is to translate the first two moments of the variance estimator into an effective degree of freedom. That degree of freedom then enters the final

Student reference.

The separate mechanism audit supports this use with qualifications. An oracle scaled chi-squared distribution fitted the complete distribution about as well as an oracle normal model and better than an oracle lognormal model. It produced the smallest upper-tail error among the three candidates. The larger discrepancy arose in the first-order prediction of finite-sample moments, not in the choice between chi-squared and normal shape families. The plug-in component degrees of freedom nevertheless had median ratio 0.970 to empirical moment-matched values.

These findings justify scaled chi-squared moment matching as a useful approximation for tail calibration. They do not establish an exact pivot. The finite-sample performance demonstrated in Chapter 6 remains empirical evidence for a theoretically motivated approximation.

7.4 Relation to existing methods

The expanded procedure occupies a middle position between normal Wald inference and resampling. Like Wald inference, it is deterministic, analytic, and linear in the number of cells. Like Welch–Satterthwaite inference, it acknowledges that an estimated denominator changes the reference tails. Its distinctive step is to calculate that denominator uncertainty for the MI variance functional rather than importing the ordinary sample-variance degrees of freedom.

The method also differs from an independence test. Pearson and likelihood-ratio tests compare a single joint distribution with the product of its own margins and use a multivariate quadratic statistic. The present method compares one scalar functional across two independent populations and uses a signed, studentized difference. At the independence boundary, the first-order MI influence vanishes and the regular derivation used here no longer applies. Appendix D develops this distinction.

The literature review identified prior ingredients rather than a direct precedent for the complete method. Bias correction for information measures, first-order MI variance, influence-function analysis, tests for entropy-based indices, and Welch–Satterthwaite moment matching are established. The claim of this thesis is consequently methodological and specific: it constructs and evaluates an MI-variance influence-function degree-of-freedom correction for the two-sample equal-MI problem. It does not claim invention of the component theories.

7.5 Limitations

Several limitations define the strength of the conclusions.

First, the primary evidence is simulation-based. The 60 fixed population pairs cover multiple table sizes, marginal shapes, association structures, sample ratios, and expected-count regimes, but they do not span all discrete distributions. A new independently generated confirmatory grid would provide a stronger guard against conclusions that depend on the chosen population generator.

Second, the theory assumes fixed finite alphabets, positive population support, independent samples, and positive first-order MI variance. The primary experiment deliberately uses $I(P) = I(Q) > 0$. It does not validate the method at independence, where the first derivative is zero, or when the alphabet grows with sample size.

Third, every analytic method uses the same first-order standard error and the same Miller-type bias correction. Expanded Welch changes the reference distribution only. Residual errors can therefore arise from finite-sample bias in the numerator, error in the first-order variance estimate, dependence between numerator and denominator, or discreteness of the table distribution. The experiment does not identify one of these as the sole remaining cause.

Fourth, invalid replicates are conditioned out uniformly across methods. This is appropriate for comparing references on common inputs, but a low valid rate would limit practical use regardless of conditional calibration. The widespread-sparsity result illustrates this point.

Fifth, the power experiment contains five controlled alternatives rather than a broad power surface. It establishes the approximate cost of the heavier reference in representative 3×3 cases, but does not characterise power over all effect sizes, table shapes, or sample ratios.

Finally, no real-data case study is included. The thesis establishes a method and its controlled numerical behaviour. Applied work is needed to determine how often its difficult regimes arise in specific fields and whether its calibration advantage changes scientific conclusions.

7.6 Recommended operating range

The results support expanded Welch when two independent categorical samples have positive estimated MI and the analyst seeks a fast analytic comparison of their population MI values. It is most useful when one or both tables have skewed margins, rare expected cells, or unequal sample sizes, but retain enough observed

support for positive first-order variance estimates.

Normal Wald remains adequate in well-sampled tables and is simpler. Expanded Welch should not be presented as a repair for arbitrary sparsity. If complete rows or columns are frequently absent, if the estimated first-order variance is zero, or if the scientific null is independence, the regular two-sample method is not the appropriate reference without further development.

7.7 Further work

The first priority is an independently generated confirmatory experiment with the analysis code and decision rules fixed in advance. It should preserve the current regime definitions, report scenario-level uncertainty, and include a broader power surface. This would test whether the observed calibration gains generalise beyond the development grid.

The second priority is improved finite-sample centring. The expanded method models uncertainty in the denominator but does not account for all distribution-specific bias in the difference. An analytically tractable higher-order correction could address the remaining liberal behaviour without introducing resampling.

The third priority is boundary theory. Near $I = 0$, first-order normal inference transitions to the second-order chi-squared geometry of independence testing. A unified method would require an explicit treatment of that transition rather than a direct extension of the present degrees of freedom.

Finally, the same variance-functional strategy may be investigated for conditional MI and related discrete information measures. Such extensions should follow, rather than precede, independent validation of the simpler two-sample MI setting.

Chapter 8

Conclusion

This thesis developed a deterministic finite-sample reference for comparing mutual information between two independent categorical populations. The method begins with the same bias-corrected MI difference and first-order standard error as normal Wald inference. It then estimates how much each population’s MI variance estimate would change across repeated samples, converts that uncertainty into component degrees of freedom by Satterthwaite moment matching, and combines the components in the usual Welch form.

The resulting expanded Welch–Satterthwaite test is mathematically grounded in the influence function of the complete MI variance functional. It accounts for changes in both the joint probabilities and the row and column margins. Its computational complexity is $O(rc)$, matching the order of the normal Wald and simple Welch alternatives.

Across 60 fixed equal-MI population pairs and 600,000 null replicates, expanded Welch had the smallest overall calibration error and the coverage closest to 95%. At $\alpha = 0.05$, mean absolute false-positive-rate error fell from 0.01342 for normal Wald and 0.01194 for simple Welch to 0.00825. The largest improvements occurred under skewness, rare expected cells, and unequal sample sizes. In well-sampled tables, expanded Welch and normal Wald were similarly accurate. The correction cost approximately one percentage point of power in the tested alternatives and increased median runtime from about 0.08 to 0.15 milliseconds per table pair.

The result is useful but bounded. Expanded Welch improves the calibration of regular positive-MI comparisons; it is not an exact finite-sample test and does not solve the nonregular independence boundary or arbitrary support loss. The most extreme sparse and imbalanced configuration remained liberal.

The principal conclusion is therefore that MI-specific Welch–Satterthwaite calibration is a viable analytic refinement for the two-sample equal-MI problem. It

offers a practical middle ground between an uncorrected normal reference and computational resampling. Independent confirmatory validation and higher-order centring are the next steps required before broad applied adoption.

Appendix A

Additional Derivation Details

This appendix records algebra that supports Chapters 3 and 4 but is not required to follow the main argument.

A.1 Gradient of mutual information on the probability simplex

Mutual information can be written as

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (\text{A.1})$$

Treating the cell probabilities as coordinates gives

$$\frac{\partial I(P)}{\partial p_{ij}} = (\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \quad (\text{A.2})$$

$$= \log \frac{p_{ij}}{p_{i+} p_{+j}} - 1 \quad (\text{A.3})$$

$$= \ell_P(i, j) - 1. \quad (\text{A.4})$$

Any valid probability perturbation $h = (h_{ij})$ satisfies $\sum_{i,j} h_{ij} = 0$. Its directional derivative is therefore

$$\left. \frac{d}{d\varepsilon} I(P + \varepsilon h) \right|_{\varepsilon=0} = \sum_{i,j} \{\ell_P(i, j) - 1\} h_{ij} \quad (\text{A.5})$$

$$= \sum_{i,j} \ell_P(i, j) h_{ij}. \quad (\text{A.6})$$

The constant in the unconstrained gradient disappears on the simplex. For the one-cell direction $h_{ij} = \mathbf{1}\{i = x, j = y\} - p_{ij}$, Equation (A.6) becomes

$$\ell_P(x, y) - \sum_{i,j} p_{ij} \ell_P(i, j) = \ell_P(x, y) - I(P), \quad (\text{A.7})$$

which is Equation (4.19).

A.2 Why the variance of pointwise MI appears

For empirical cell probabilities,

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\}. \quad (\text{A.8})$$

Substitution into the first-order expansion of MI gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i, j) (\hat{p}_{ij} - p_{ij}) \quad (\text{A.9})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \left[\sum_{i,j} \ell_P(i, j) \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\} \right] \quad (\text{A.10})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \{\ell_P(Z_a^{(P)}) - I(P)\}. \quad (\text{A.11})$$

The observation-level summand has variance

$$\text{Var}_P\{\ell_P(X, Y) - I(P)\} = \mathbb{E}_P \left[\{\ell_P(X, Y) - I(P)\}^2 \right] \quad (\text{A.12})$$

$$= \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2 \quad (\text{A.13})$$

$$= V(P). \quad (\text{A.14})$$

This is why the first-order variance of the complete MI estimator is $V(P)/n_P$, rather than a variance of the numerical category labels.

A.3 Mean of the variance influence function

The influence function $g_P(x, y)$ is centred at the population level. This can be shown without expanding its individual terms. The weighted average of the one-cell

perturbation directions is

$$\sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\} = P - P = 0. \quad (\text{A.15})$$

Directional differentiation is linear in its direction. Hence

$$\mathbb{E}_P\{g_P(X, Y)\} = \sum_{x,y} p_{xy} \left. \frac{d}{d\varepsilon} V\{(1 - \varepsilon)P + \varepsilon\delta_{(x,y)}\} \right|_0 \quad (\text{A.16})$$

$$= DV_P \left[\sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\} \right] \quad (\text{A.17})$$

$$= DV_P[0] = 0. \quad (\text{A.18})$$

The empirical implementation nevertheless subtracts $\bar{g}_P$ in Equation (4.38). This removes floating-point and plug-in centring error without changing the population definition.

A.4 Direct table substitutions

For observed counts $N^P = (N_{ij}^P)$, the quantities used by expanded Welch are calculated as

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{p}_{i+} = \sum_j \hat{p}_{ij}, \quad \hat{p}_{+j} = \sum_i \hat{p}_{ij}, \quad (\text{A.19})$$

$$\hat{\ell}_P(i, j) = \log \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}}, \quad \hat{I}(P) = \sum_{i,j} \hat{p}_{ij} \hat{\ell}_P(i, j), \quad (\text{A.20})$$

$$\hat{V}(P) = \sum_{i,j} \hat{p}_{ij} \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2. \quad (\text{A.21})$$

All sums involving pointwise MI are restricted to cells with $\hat{p}_{ij} > 0$. The probability-weighted row and column means are

$$\hat{R}_P(i) = \frac{\sum_j \hat{p}_{ij} \hat{\ell}_P(i, j)}{\hat{p}_{i+}}, \quad (\text{A.22})$$

$$\hat{C}_P(j) = \frac{\sum_i \hat{p}_{ij} \hat{\ell}_P(i, j)}{\hat{p}_{+j}}. \quad (\text{A.23})$$

Consequently,

$$\hat{g}_P(i, j) = \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \quad (\text{A.24})$$

$$+ 2\{\hat{\ell}_P(i, j) - \hat{R}_P(i) - \hat{C}_P(j) + \hat{I}(P)\}, \quad (\text{A.25})$$

$$\bar{g}_P = \sum_{i,j} \hat{p}_{ij} \hat{g}_P(i, j), \quad (\text{A.26})$$

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \{\hat{g}_P(i, j) - \bar{g}_P\}^2, \quad (\text{A.27})$$

$$\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}. \quad (\text{A.28})$$

These expressions are the direct $O(rc)$ implementation of the population derivation.

A.5 Numerical derivative checks

The implementation tests the analytic derivatives against centred finite differences on positive probability tables. For a selected cell, define

$$P_{\pm h} = (1 \mp h)P \pm h\delta_{(x,y)}. \quad (\text{A.29})$$

For sufficiently small h , the checks compare

$$\frac{I(P_h) - I(P_{-h})}{2h} \quad \text{with} \quad \ell_P(x, y) - I(P), \quad (\text{A.30})$$

$$\frac{V(P_h) - V(P_{-h})}{2h} \quad \text{with} \quad g_P(x, y). \quad (\text{A.31})$$

The finite-difference step is selected so that both perturbed tables retain positive probabilities. These checks test the algebra independently of the Monte Carlo calibration experiment.

Appendix B

Complete Scenario Specifications

Table B.1 records the realised sample sizes and minimum true joint expected counts for every fixed population pair in the primary calibration experiment. The population tables themselves contain many cell probabilities and are retained without rounding in `population_scenarios.csv`. That file is the authoritative machine-readable specification.

In the table, $E_{\min}^P = \min_{i,j} n_P p_{ij}$ and $E_{\min}^Q = \min_{i,j} n_Q q_{ij}$. Every row represents 10,000 independently sampled table pairs under the exact equal-MI null. The largest numerical departure from $I(P) = I(Q)$ was 8.5×10^{-14} nats.

Table B.1: Realised design of all 60 equal-MI population pairs.

Shape	Regime variant	n_P	n_Q	$E_{\min}^P$	$E_{\min}^Q$
(2×2)	Well sampled 1	200	200	18.228	19.953
(2×2)	Well sampled 2	200	200	15.015	19.561
(2×2)	Moderate 1	100	100	10.697	6.927
(2×2)	Moderate 2	100	200	4.427	0.942
(2×2)	Highly skewed and sparse 1	52	52	4.660	4.072
(2×2)	Highly skewed and sparse 2	54	108	4.201	4.622
(2×2)	Ultra-skewed and sparse 1	142	142	0.961	0.052
(2×2)	Ultra-skewed and sparse 2	51	510	0.964	0.073
(2×2)	Widespread sparsity 1	50	50	0.042	0.048
(2×2)	Widespread sparsity 2	50	100	0.506	0.312
(2×5)	Well sampled 1	200	200	5.597	6.526

Continued on next page

Table B.1 continued

Shape	Regime variant	n_P	n_Q	$E_{\min}^P$	$E_{\min}^Q$
(2×5)	Well sampled 2	200	200	1.385	4.203
(2×5)	Moderate 1	100	100	1.702	1.738
(2×5)	Moderate 2	100	200	1.656	0.528
(2×5)	Highly skewed and sparse 1	190	190	1.575	3.653
(2×5)	Highly skewed and sparse 2	158	316	3.580	1.632
(2×5)	Ultra-skewed and sparse 1	121	121	0.630	0.129
(2×5)	Ultra-skewed and sparse 2	69	690	0.771	0.632
(2×5)	Widespread sparsity 1	50	50	(3.33×10^{-6})	0.038
(2×5)	Widespread sparsity 2	50	100	(1.09×10^{-6})	0.011
(3×3)	Well sampled 1	200	200	9.084	6.111
(3×3)	Well sampled 2	200	200	7.214	3.666
(3×3)	Moderate 1	100	100	1.758	2.727
(3×3)	Moderate 2	100	200	1.178	3.135
(3×3)	Highly skewed and sparse 1	277	277	1.365	3.940
(3×3)	Highly skewed and sparse 2	433	866	4.177	1.181
(3×3)	Ultra-skewed and sparse 1	64	64	0.808	0.168
(3×3)	Ultra-skewed and sparse 2	70	700	0.768	0.610
(3×3)	Widespread sparsity 1	50	50	0.268	(5.11×10^{-4})
(3×3)	Widespread sparsity 2	59	118	(4.20×10^{-4})	0.002
(3×5)	Well sampled 1	225	225	2.442	2.507
(3×5)	Well sampled 2	225	225	3.597	5.426
(3×5)	Moderate 1	120	120	1.752	2.675
(3×5)	Moderate 2	100	200	0.437	2.007
(3×5)	Highly skewed and sparse 1	414	414	2.086	3.290
(3×5)	Highly skewed and sparse 2	337	674	1.460	3.797
(3×5)	Ultra-skewed and sparse 1	535	535	0.711	0.084
(3×5)	Ultra-skewed and sparse 2	69	690	0.881	0.057
(3×5)	Widespread sparsity 1	55	55	(3.47×10^{-4})	0.216
(3×5)	Widespread sparsity 2	87	174	0.147	(5.34×10^{-10})

Continued on next page

Table B.1 continued

Shape	Regime variant	n_P	n_Q	$E_{\min}^P$	$E_{\min}^Q$
(5×5)	Well sampled 1	375	375	4.830	2.140
(5×5)	Well sampled 2	375	375	1.923	2.192
(5×5)	Moderate 1	200	200	1.487	0.820
(5×5)	Moderate 2	150	300	0.474	2.101
(5×5)	Highly skewed and sparse 1	619	619	2.591	2.768
(5×5)	Highly skewed and sparse 2	297	594	1.544	3.700
(5×5)	Ultra-skewed and sparse 1	473	473	0.413	0.533
(5×5)	Ultra-skewed and sparse 2	75	750	0.304	0.139
(5×5)	Widespread sparsity 1	435	435	(6.06×10^{-6})	0.025
(5×5)	Widespread sparsity 2	414	828	0.015	(3.28×10^{-6})
(8×8)	Well sampled 1	960	960	3.464	4.233
(8×8)	Well sampled 2	960	960	2.379	2.341
(8×8)	Moderate 1	512	512	1.525	0.790
(8×8)	Moderate 2	384	768	0.351	0.057
(8×8)	Highly skewed and sparse 1	975	975	1.373	1.025
(8×8)	Highly skewed and sparse 2	460	920	1.441	1.096
(8×8)	Ultra-skewed and sparse 1	604	604	0.145	0.213
(8×8)	Ultra-skewed and sparse 2	75	750	0.255	0.212
(8×8)	Widespread sparsity 1	417	417	(6.28×10^{-5})	0.003
(8×8)	Widespread sparsity 2	271	542	0.003	0.015

The sparse sample sizes were selected from the true population probabilities to satisfy the predeclared expected-count constraints. They were fixed before replicate tables were sampled. No replicate-level outcome was used to choose its sample size, population, or analysis method.

Appendix C

Additional Validation

C.1 Scaled chi-squared audit design

The component degree of freedom in Equation (4.39) is based on a scaled chi-squared working model for the estimated MI variance. A separate simulation isolated this mechanism from the final two-sample test. It used 64 fixed scenarios and evaluated both population components, giving 128 component distributions. For each component, 10,000 multinomial tables estimated oracle moments and an independent 10,000 tables evaluated fit.

The audit made two comparisons. The oracle comparison supplied the empirical mean and variance to scaled chi-squared, normal, and lognormal distributions. Differences between these models therefore measure distributional shape. The population first-order comparison instead used $V(P)$ and $\tau^2(P)/n_P$ from the derivation, so it includes error in the predicted moments.

Kolmogorov–Smirnov (KS) distance measures whole-distribution disagreement. Tail error is the absolute difference between the observed exceedance probability at a candidate quantile and its nominal probability.

Table C.1: Overall distributional audit for the MI variance estimator.

Model	Mean KS	Median KS	Tail error .05	Tail error .01	Win rate
Oracle scaled chi-squared	0.0199	0.0143	0.0045	0.0026	0.4219
Oracle normal	0.0196	0.0143	0.0054	0.0038	0.4688
Oracle lognormal	0.0291	0.0227	0.0062	0.0040	0.1094
Population first-order chi-squared	0.1622	0.0822	0.0765	0.0345	–

The oracle normal model had the smallest mean KS distance by 0.0003. The scaled chi-squared model had smaller upper-tail errors at both evaluated levels. The chi-squared working family is therefore competitive for the part of the distribution

that determines two-sided rejection, although it is not uniformly the closest family.

C.2 Audit by table shape

Table C.2: Oracle distributional fit by table shape.

Shape	Chi KS	Normal KS	Chi tail .05	Normal tail .05	Empirical skew	DF ratio
(2×2)	0.0354	0.0273	0.0080	0.0075	0.1920	0.9895
(3×3)	0.0209	0.0197	0.0056	0.0058	0.2095	0.9695
(5×5)	0.0128	0.0176	0.0026	0.0044	0.1827	0.9658
(10×10)	0.0105	0.0137	0.0017	0.0038	0.1071	0.9568

The scaled chi-squared shape becomes more competitive as table dimension increases. The final column is the ratio of the plug-in component degree of freedom to the empirical moment-matched degree of freedom. The median ratio over all 128 components was 0.9703.

C.3 Audit by sampling regime

Table C.3: First-order and oracle scaled chi-squared diagnostics by regime.

Regime	Components	DF ratio	Oracle KS	First-order KS	Tail .05
Well sampled	16	0.9585	0.0166	0.2172	0.0960
Moderate	16	0.9546	0.0231	0.2065	0.0963
Sparse and imbalanced	16	0.9649	0.0188	0.1488	0.0622
Highly skewed and sparse	16	0.9946	0.0164	0.0462	0.0128
Ultra-skewed and sparse	16	1.0036	0.0173	0.0786	0.0150
Widespread sparsity	16	0.8522	0.0272	0.3081	0.2042
Equal-MI shape mismatch	16	0.9658	0.0236	0.1437	0.0609
Extreme sample imbalance	16	0.9881	0.0161	0.1483	0.0648

The largest failure occurs under widespread sparsity. The oracle chi-squared fit also deteriorates there, and first-order moment error becomes substantial. This agrees with the lower valid rate and residual calibration error in the main experiment.

C.4 Interpretive limit

This component audit does not make the final Student reference exact. The MI contrast and its estimated standard error are computed from the same tables and can

remain dependent. The audit establishes that scaled chi-squared moment matching is a plausible model for the positive variance estimator, especially in the upper tail, while identifying the first-order moments as a larger source of finite-sample error.

Appendix D

The Independence Boundary

The main thesis concerns the regular weak null ($I(P)=I(Q)>0$). This appendix explains why constructing an independent reference distribution does not turn the expanded Welch method into a new first-order test of independence.

D.1 Constructing the independent reference

For a joint distribution $P = (p_{ij})$, define

$$q_{ij} = p_{i+}p_{+j}. \quad (\text{D.1})$$

Then $Q = (q_{ij})$ is a probability distribution because

$$\sum_{i,j} q_{ij} = \left(\sum_i p_{i+} \right) \left(\sum_j p_{+j} \right) = 1. \quad (\text{D.2})$$

It has the same margins as (P) , since

$$\sum_j q_{ij} = p_{i+}, \quad \sum_i q_{ij} = p_{+j}. \quad (\text{D.3})$$

The factorisation in Equation (D.1) makes (X) and (Y) independent under (Q) , so ($I(Q)=0$).

The divergence from (P) to this reference is

$$D_{\text{KL}}(P||Q) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (\text{D.4})$$

$$= \sum_{i,j} p_{ij} \log \frac{p_{ij}}{p_{i+}p_{+j}} \quad (\text{D.5})$$

$$= I(P). \quad (\text{D.6})$$

Consequently, testing $(I(P)=I(Q))$ for this constructed reference is exactly testing $(I(P)=0)$, or independence.

D.2 Why the first-order term vanishes

At independence, $p_{ij} = p_{i+}p_{+j}$, and every population pointwise-MI value is zero:

$$\ell_P(i, j) = \log 1 = 0. \quad (\text{D.7})$$

Equation (4.19) therefore gives

$$\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P) = 0. \quad (\text{D.8})$$

The first-order variance also vanishes:

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\} = 0. \quad (\text{D.9})$$

The denominator used by normal Wald, simple Welch, and expanded Welch is thus degenerate under the population independence null. Changing its degrees of freedom cannot repair a statistic whose leading term is zero.

D.3 Second-order behaviour

Let an empirical perturbation around an independent distribution have first-order cell residuals h_{ij} , with row and column sums h_{i+} and h_{+j} . The association component is

$$a_{ij} = h_{ij} - p_{+j}h_{i+} - p_{i+}h_{+j}. \quad (\text{D.10})$$

The linear contribution to MI is zero. Expanding the logarithm to second order gives

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \sum_{i,j} \frac{a_{ij}^2}{p_{i+}p_{+j}} + o(\varepsilon^2). \quad (\text{D.11})$$

For multinomial sampling, empirical probability errors are of order $n^{-1/2}$. Equation (D.11) is therefore of order n^{-1} , and the nondegenerate statistic is $2n\hat{I}(P)$, not $\sqrt{n}\{\hat{I}(P) - I(P)\}$. Its regular fixed-alphabet limit is

$$2n\hat{I}(P) \xrightarrow{d} \chi_{(r-1)(c-1)}^2. \quad (\text{D.12})$$

This is the usual likelihood-ratio test of independence. The chi-squared reference arises from the second-order quadratic geometry at the boundary, whereas the expanded Welch reference arises from first-order variation away from that boundary.

D.4 What alternative constructions produce

Using the observed margins to set $\hat{q}_{ij} = \hat{p}_{i+}\hat{p}_{+j}$ gives

$$2nD_{\text{KL}}(\hat{P}\|\hat{Q}) = 2n\hat{I}(P), \quad (\text{D.13})$$

which is the likelihood-ratio statistic above.

Repeatedly shuffling one variable preserves its observations while breaking the pairing with the other variable. The resulting empirical distribution is a permutation reference. Repeatedly drawing tables from $\hat{P}_X\hat{P}_Y$ is a parametric bootstrap reference. A single simulated Q table adds Monte Carlo variation but does not by itself define a calibrated null distribution.

The independent reference $Q = P_X P_Y$ is therefore mathematically valid, but it does not create a regular two-sample MI comparison. Analytic improvement of the sparse independence test would require a second-order finite-sample method, rather than an immediate extension of the expanded Welch calculation.

Appendix E

Reproducibility

E.1 Software and random seeds

The primary experiment was run with Python 3.13.5, NumPy 2.4.4, SciPy 1.17.1, and pandas 3.0.3. Its population-generation seed was 2,026,080,501 and its simulation seed was 2,026,080,502. The scaled chi-squared audit used seeds 2,026,080,701 and 2,026,080,702. The exact settings and software versions are stored beside each output in `run_metadata.json`.

E.2 Primary experiment

From the repository root, the primary experiment is reproduced with

```
MPLBACKEND=Agg MPLCONFIGDIR=$PWD/.mplcache \  
  .venv/bin/python \  
  WelchSatterthwaiteMI/experiments/run_supervisor_experiment.py \  
  --profile full \  
  --output-dir WelchSatterthwaiteMI/results/supervisor_practical
```

The full profile uses 10,000 null replicates per scenario, 10,000 replicates per power setting, batches of 1,000, and 200 timing repetitions. The recorded SHA-256 hash of the experiment script was

9e654a849ff100eedd57c2703ee5617a32d7c35b3400ef37be718adca35f9123

The output directory contains the following principal artefacts:

- `population_scenarios.csv`:
Fixed population probabilities, sample sizes, generating parameters, and expected-count diagnostics for all 60 null scenarios.

- `scenario_results.csv`:
Scenario-level FPRs, calibration errors, Wilson intervals, coverage, and validity diagnostics.
- `regime_summary.csv`:
Equal-weight summaries over the 12 population pairs in each regime.
- `null_pvalues.npz`:
Replicate-level p-values used to construct the rejection-calibration curves.
- Calibration-grid files:
`rejection_calibration_scenarios.csv` and
`rejection_calibration_regimes.csv`.
Actual rejection rates over the complete nominal-alpha grid.
- `power_summary.csv` and `runtime_summary.csv`:
Alternative-hypothesis and computational results.

E.3 Scaled chi-squared audit

The component audit is reproduced with

```
MPLBACKEND=Agg MPLCONFIGDIR=$PWD/.mplcache \
  .venv/bin/python \
  WelchSatterthwaiteMI/experiments/validate_scaled_chi_square.py \
  --profile focused \
  --replicates 10000 \
  --output-dir \
  WelchSatterthwaiteMI/results/scaled_chi_square_validation
```

The corresponding output directory contains component-, regime-, shape-, and model-level summaries together with the fixed populations and metadata.

E.4 Automated checks

The implementation tests are run with

```
.venv/bin/python -m unittest discover \
  -s WelchSatterthwaiteMI/tests -v
```

The tests cover MI calculation, bias correction, normal and Welch references, the variance influence calculation, table validation, reproducibility of the experiment profiles, and scaled chi-squared audit utilities.

E.5 Building this document

The scenario-calibration, effective-degrees-of-freedom, and power figures are generated from the accepted CSV outputs. From `Thesis Writeup/Welch MI`, run

```
MPLBACKEND=Agg MPLCONFIGDIR=/tmp/welch-mi-mpl \  
  ../../.venv/bin/python figures/make_thesis_figures.py
```

The thesis is then built with

```
/Library/TeX/texbin/latexmk -pdf \  
  -interaction=nonstopmode -halt-on-error main.tex
```

The bibliography is read from `WelchSatterthwaiteMI/article/references.bib`. The calibration figure and all thesis-specific summary figures are generated from the accepted experiment outputs rather than from separately edited values.

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